

Representation in Product Development: Evidence from Insurance and Clinical Trials*

Jennifer Kao
UCLA Anderson

Tamar Oostrom
The Ohio State University

March 25, 2025

Abstract

We investigate the causes and consequences of demographic disparities in product development. In 2000, Medicare expanded coverage of clinical trial costs, reducing participation costs for elderly enrollees. This policy increased trials for diseases common among the elderly by 24 percent, led trial sponsors to broaden enrollment criteria, and may have increased elderly enrollment. This policy was associated with increased prescription utilization among elderly individuals. To explore potential mechanisms, we examine clinical outcomes and adverse events. Our findings highlight how financial incentives can shape research priorities and improve representation in research.

JEL Classifications: O31, O38, I14, I18, J14

Keywords: Inequality, Innovation and Invention, Health Policy, Economics of the Elderly

*We are grateful to seminar participants at the Berkeley Health Economics Workshop, Electronic Health Economics Colloquium, HBS Junior Innovation Economics Conference, Junior(ish) Health Economics Summit, NBER Productivity, Innovation, and Entrepreneurship Program Meeting, Toulouse School of Economics, UCLA, University of Kansas, University of Toronto, USC, and Washington State University for helpful feedback on this project. We thank clinical research finance experts at UCLA for helpful discussions. We thank Grace Nussle for excellent research assistance. All errors are our own.

1 Introduction

During the research and development (R&D) process, firms must decide which users to engage to develop and test their products. Diverse representation may support the development of products that effectively target these users and encourage adoption.¹ However, increasing diversity in R&D may entail additional costs and complex logistics. In this paper, we study how financial barriers shape user representation in R&D and examine how changes in representation impact innovation, product adoption, and quality.

We focus on drug development, where firms enroll participants in clinical trials to generate information about a drug’s safety and efficacy. This information shapes the decisions of regulators, physicians, and patients. In recent years, policymakers and researchers have called for governments to address the persistent underrepresentation of specific demographic groups—specifically, the elderly, women, and Black individuals—in clinical trials relative to the prevalence of the diseases affecting these groups. For example, elderly individuals make up nearly 63 percent of new cancer cases, but only comprise 25 percent of participants in cancer clinical trials (Hutchins et al., 1999). While trust (Alsan and Wanamaker, 2017), stereotypes (Hebert, 2019), and efficacy and safety concerns (Nadel, 1992) may contribute to participant disparities, financial barriers are frequently cited as a significant driver (Lara et al., 2001).

This paper investigates how expanding insurance coverage of clinical trials for elderly individuals affects innovation in elderly diseases, representation in clinical trials, prescription utilization, and quality. In 2000, President Clinton passed an executive memorandum requiring Medicare to cover the routine patient costs associated with clinical trial participation. This policy was enacted in response to concerns that low clinical trial representation of participants aged over 65 may lead to delays in trial enrollment and higher drug development costs. Prior to 2000, drug manufacturers or participants were often responsible for the high routine patient costs in clinical trials.

To assess the Medicare Memorandum’s impact, we use a difference-in-differences (DID) design exploiting cross-disease variation in policy exposure. We examine (1) the number of clinical trials for diseases prevalent in the elderly (aged over 65), (2) the number of trials that broadened their

¹“Representation” is defined as the extent to which the testing users’ demographics mirror the composition of the eventual end-users.

enrollment criteria to include elderly participants, (3) the share of elderly participants enrolled, (4) utilization of pharmaceuticals among the elderly, and (5) measures of quality, such as trial precision and adverse events. We assemble a new dataset of clinical trials, including the trial start and end date, the diseases under investigation, participant enrollment criteria, and actual enrollment. We link this dataset to detailed information on drug utilization, clinical trial outcomes, and adverse events.

This policy meaningfully shifted drug development. Diseases most affected by the memorandum experienced a 24 percent increase in the number of Phase II and Phase III clinical trials, relative to diseases that were least affected. We find an immediate increase in innovation largely driven by drugs that had previously undergone clinical testing. These effects are larger in diseases with a high cost for clinical trials and larger in the United States compared to Europe.

Our analysis spans 1995 to 2005, when Medicare did not cover prescription drug costs outside of those administered in a physician’s office or hospital. Medicare Part D, which expanded prescription drug coverage for Medicare recipients, was signed in December 2003 and implemented in January 2006. Importantly, we find that the increase in research among elderly diseases occurs immediately after the memorandum, before Medicare Part D was signed into law or implemented, and we exclude years after its introduction in 2006.²

Next, we investigate whether this increase in elderly-focused innovation affected trial design. In elderly diseases, firms increased the number of clinical trials that allowed elderly enrollment by 26 percent, with no change in trials that allowed non-elderly participants. We find noisy but suggestive evidence that this change in enrollment criteria led to an increase in the share of elderly enrollees.

The benefits or costs of increased representation are most compelling if they shape product utilization and patient outcomes. The representation of elderly trial participants has the potential to shape utilization, due to the dissemination of clinical trial information through drug labels, public registries, publications, and advertisements. We find that following the memorandum, there was a disproportionate increase in drug utilization among elderly participants for high Medicare share diseases, relative to both low Medicare share diseases and non-elderly participants. Similar shifts

²The passage of prescription drug coverage for Medicare had been debated for over two decades and the passage of the final bill was a surprise with thin margins (Alpert, 2016).

in utilization occurred among individuals likely to already have drug insurance who would be less affected by the anticipation of Medicare Part D. This approach uses variation across diseases in exposure to the memorandum and variation in the ages of individuals using these pharmaceuticals.

These findings are economically meaningful: we observe an average of 1.09 Phase II and III trials within a disease each year. [Sertkaya et al. \(2016\)](#) estimate that the average cost of a clinical trial is \$16.62 million, suggesting that the 24 percent increase due to the Medicare Memorandum of 2000 spurred clinical investments of \$4.4 million annually per disease common among the elderly. Importantly, these investments led to a significant increase in targeting and enrollment of elderly individuals.

Our paper contributes to the literature on representation in research and innovation (e.g. [Green et al. \(2022\)](#) and [Hutchins et al. \(1999\)](#)). First, we provide causal, large-scale evidence of how financial incentives shape representation in R&D. Our work is related to [Michelman and Msall \(2022\)](#) who study the effect of regulatory guidance on including women in clinical trials. We observe enrollment criteria and actual enrollment and find effects on clinical trials, suggesting that direct financial incentives may be more impactful than regulatory guidance. Recent work shows that gender and race-concordance of investigators are associated with more participants and interest among that demographic group ([Gupta, 2022](#); [Alsan et al., 2025](#)) but does not investigate the causal effects of a policy on representation.

Second, we show that representation in R&D influences product adoption. Recent work by [Alsan et al. \(2024\)](#) provides survey evidence that representation affects utilization for Black Americans. Previous research has found that prescription decisions reflect both clinical trials and future regulatory decisions ([Ristovska, 2025](#); [McKibbin, 2023](#)). We provide evidence that representation matters in real-world prescription data.

Other related work has shown that inventor representation affects project development and firm outcomes ([Nielsen et al., 2017](#); [Cao, Koning and Nanda, 2023](#); [Koning, Samila and Ferguson, 2021](#)). We focus on the effects of funding—a key input to innovation—rather than innovators.

Finally, we highlight how public health insurance expansions shape innovation. Previous studies have examined how insurance affects pharmaceutical innovation by shifting expected demand (e.g., [Blume-Kohout and Sood \(2013\)](#)). To our knowledge, ours is the first to examine how expanding

insurance coverage affects the supply of R&D inputs (i.e., user groups) and pharmaceutical innovation. We also contribute to a literature outside economics on the Medicare Memorandum of 2000. This literature uses subsets of diseases or cross-sectional evidence (Unger et al., 2006). We provide the first comprehensive, econometrically-based analysis of the memorandum.

While we do not fully evaluate the policy’s overall welfare effects, the observed positive medium-term impact on the number of trials and drug utilization suggests that reducing financial barriers to participant enrollment could be effective for shaping innovation and increasing representation.

2 Institutional Background

2.1 Drug Development

In the US, drug development involves clinical trials with rising costs at each phase. Drugs that successfully demonstrate safety in Phase I trials proceed to Phase II trials in which their efficacy is tested in a few hundred participants. If successful, the drugs move to Phase III trials in which their efficacy is tested in several thousand participants. Upon successfully completing multiple Phase III trials, the sponsor can submit a New Drug Application to the FDA for final approval.

2.2 Costs of Clinical Research

The drug development process is costly (typically costing a drug manufacturer \$800 million) and long (often taking between 8 and 12 years) (DiMasi, Hansen and Grabowski, 2003; DiMasi, Grabowski and Hansen, 2016). A key driver of these costs is participant recruitment and retention costs. Estimates suggest average patient recruitment costs are \$37,000 per patient (for Phase I) to \$308,000 per patient (for Phase III) (Sertkaya et al., 2016). Drug manufacturers aim to enroll eligible participants quickly, due to competition in participant recruitment and to minimize the high fixed costs associated with extended trial recruitment periods. An average Phase III trial with direct costs of \$11 million is estimated to have \$308,000 in patient recruitment costs, \$25,000 in patient retention costs, \$2,252,000 for clinical procedures for patients, and \$850,000 for lab costs for patients (Sertkaya et al., 2014).

Before 2000, the costs of participant recruitment and retention were often shared between drug

manufacturers and participants (Pear, 2000). While there is limited information on the actual allocation of costs and substantial variation, drug manufacturers typically covered routine clinical trial expenses and data analysis (Aaron and Gelbrand, 2000). Routine costs account for the bulk of the cost of most clinical trials and include room and board for hospitalized patients, diagnostic and laboratory tests, post-surgical care, and office visits (Aaron and Gelbrand, 2000). Participants often received treatments at minimal cost but faced expenses for some medical care, screening, and travel. Medicare occasionally covered some trial costs but frequently deemed trials experimental and denied coverage. Before 2000, Medicare did not cover routine patient costs. In 2000, about 17 percent of Medicare patients had a Medicare Advantage plan, and private insurers also historically did not cover these costs.

The uncertainty surrounding the financial responsibility associated with clinical trials deterred participation and led to unexpected costs for enrollees (Pear, 2000). Clinical trial investigators could not assure elderly prospective participants that Medicare would cover trial costs, a risk that was required to be disclosed during the informed consent process (Aaron and Gelbrand, 2000). Additionally, Medicare’s policy of not routinely preauthorizing care and its legal status as a payer of last resort meant trial participants incurred significant costs if Medicare denied the manufacturer’s claims. Taken together, this created substantial uncertainty for participants and manufacturers. To address some of this uncertainty, about one-third of all industry trials were fully funded and covered all routine costs without billing insurance.³

2.3 Representation of the Elderly In Clinical Trials

The high costs and uncertainty of insurance coverage may have contributed to the limited participation of elderly participants in clinical trials. Although individuals aged over 65 constitute 16.8 percent of the U.S. population, they account for just 6 percent of participants in the average pre-2000 trial in our dataset (U.S. Census Bureau, 2023). Individuals aged over 65 have been reported to be underrepresented in diseases such as cancer, diabetes, heart failure, and osteoporosis (Hutchins et al., 1999; Lau et al., 2022).

Beyond cost concerns, the elderly encounter logistics, comorbidities, health literacy, and cogni-

³Source: Author’s interviews with clinical trial managers.

tive barriers to trial participation (Herrera et al., 2010). For drug manufacturers, a homogeneous trial population reduces the variance of treatment effects, lowers the sample size needed for statistical power, and thus makes the inclusion of the elderly potentially more costly.

2.4 Medicare Memorandum of 2000

In June of 2000, President Clinton ordered Medicare to start covering most of the costs of clinical trials for its beneficiaries (largely patients aged 65 and older). The memorandum stated, “Medicare will immediately begin to reimburse for the routine patient care costs, and costs due to medical complications associated with participation in a clinical trial.” This memorandum included items or services typically provided absent a clinical trial, such as conventional care during a trial. This also includes monitoring of the effects of the investigational drug or service, items required for the provision of the investigational drug or service, and costs related to the diagnosis and treatment of complications. Drug manufacturers would continue to pay for the collection and analysis of data and the investigational item or service itself.⁴

Vice President Al Gore, speaking about the memorandum, said, “Speeding up enrollment can accelerate the discovery and use of cost-saving, life-saving, new therapies” (Pear, 2000). This memorandum also included an informational campaign to inform Medicare beneficiaries, doctors, hospitals, and other health care providers of the new policy. The intended impact of this policy was to decrease uncertainty and increase subsidization of clinical trial costs for both elderly participants and drug manufacturers. Specifically, patients were assured that their routine costs were covered and manufacturers could bill Medicare instead of risking covering costs themselves.⁵

3 Data

We are interested in measuring how the Medicare Memorandum of 2000 shifted innovation and trial enrollment for a given disease. We define diseases using the International Classification of Diseases,

⁴See National Coverage Determination (NCD) for Routine Costs in Clinical Trials (310.1)

⁵While the program primarily aimed to increase elderly participation in trials, it also sought to ensure that the trials generated information necessary for determining Medicare reimbursement for the drugs being tested.

version 9 (ICD-9).⁶ Our final dataset consists of 390 ICD-9 codes.

1. **High Medicare Share Diseases:** Following a commonly used approach (e.g., [Acemoglu et al. \(2006\)](#), [Duggan and Morton \(2010\)](#), [Blume-Kohout and Sood \(2013\)](#)), we proxy for exposure to the memorandum by calculating the share of individuals covered by Medicare associated with each ICD-9. We use the 1996 (the earliest year in which data is available) to 1999 MEPS, a nationally representative survey of the U.S. civilian non-institutionalized population ([Medical Expenditure Panel Survey, 1996-2010](#)). We categorize diseases as having a “high Medicare share” if they have more than the median share of Medicare individuals with that diagnosis from 1996-1999.⁷ Appendix Figure [A.1](#) presents the distribution of Medicare shares across diseases. For example, low Medicare share diseases (“young diseases”) include attention-deficit/hyperactivity disorder (3 percent Medicare share). In contrast, high Medicare share diseases (“old diseases”) include Parkinson’s disease (81 percent Medicare share).
2. **Clinical Trials:** To create a dataset of clinical trials and participant enrollment, we begin with data from the Cortellis Clinical Trials Intelligence Database (“Cortellis”) ([Cortellis Clinical Trials Intelligence Database, 1995-2005](#)). Cortellis contains clinical trial data from clinical trial registry websites, press releases, financial filings, and FDA submissions. This dataset contains each trial’s enrollment criteria, start year, duration, phase, results, and associated ICD-9 codes. We restrict the sample to trials that begin between 1995-2005 since data are sparse in earlier years. We restrict trials to those with at least one U.S. location, as the policy was U.S.-based.⁸ Finally, we restrict our sample to Phase II and III trials as Phase I trials are not well-reported and are arguably not covered by the memorandum.⁹ As private-sector and public-sector funders may have different objective functions, we restrict our analysis to trials with any private-sector funding. This results in a final dataset of 4,964 trial-diseases.

⁶Examples of diseases at the ICD-9 code level include “malignant neoplasm of the colon” and “diabetes mellitus.”

⁷Our results are robust to using a continuous measure as in Appendix Figure [A.2](#)

⁸We drop trials that occur in other countries, since we do not take a stand on whether international pharmaceutical investment is a complement or substitute for U.S.-based investment. We show a placebo test using European trials in Figure [A.3](#).

⁹Covered trials must evaluate efficacy, not just safety. See National Coverage Determination (NCD) for Routine Costs in Clinical Trials (310.1)

3. **Inclusion Criteria and Enrollment:** We incorporate ClinicalTrials.gov data on clinical trial enrollment characteristics, such as the number of elderly (aged 65 and over) and non-elderly participants (see Appendix Figure A.4 for an example). We supplement this data with enrollment counts from publications for a random subset of trials with a PubMed identifier. We also analyze the text of the inclusion criteria for each trial. Each clinical trial has specific enrollment criteria listing eligible participants (e.g. “Female”, “Ages 18-44”, “Type 2 diabetes”). Enrollees are required to meet these criteria. We update our enrollment counts based on the enrollment criteria when available. For example, a trial with enrollment criteria restricted to ages 18-64 should have no participants over age 65.
 4. **Drug Utilization:** We collect data on drug utilization from the 1996 to 2010 MEPS surveys. We categorize drug utilization for individuals aged 65-75 and separately for those aged 50-60. We next classify prescription drugs in the MEPS as primarily related to old or young diseases. We link MEPS drug names to clinical trials in the Cortellis data, which provides information on the diseases tested and the drug’s approval status by indication. We assign a drug to the disease for which the FDA initially approved it; we use the most common disease for that drug in Cortellis if there are several initial approvals. As above, we identify high and low Medicare share diseases based on the share of Medicare patients with that disease diagnosis. Consequently, a prescription drug in MEPS can be categorized as treating a disease with either a high or low Medicare share.
 5. **Disease Characteristics:** We characterize diseases in two ways. To determine the under and over-representation of elderly individuals across diseases, we compute each ICD-9’s average disease burden by incorporating data from The Global Burden of Disease (GBD) dataset (Global Burden of Disease, 2019). Disease burden is measured with noise, aggregated from a variety of sources and frequently unavailable so we use a 2019 cross-section instead of historical data.¹⁰
- We also estimate clinical trial costs. Because detailed ICD-9 disease-level estimates of clinical trial costs are not publicly available, we use two proxies: (1) the average trial duration and (2)

¹⁰Disease burden is an imperfect proxy of a disease’s potential market size, since market size is disease burden interacted with access to health care and willingness to pay for treatment.

whether a disease is primarily associated with outpatient or inpatient care. Longer trials are likely to incur higher costs and inpatient care is generally more costly and invasive compared to outpatient care (Cox et al., 2011; Friedrichs et al., 2016). We divide diseases by whether they are above or below the median in trial duration or prevalence in MEPS inpatient files.

We construct a panel dataset at the disease-year level. Table A.1 shows that from 1995 to 2005, the average disease-year has 1.09 total clinical trials. During our sample period, although the majority of clinical trials include elderly participants (an average of 0.93 trials per year compared to 0.15 trials per year that only had non-elderly participants), elderly individuals make up only 13% of total trial participants. To investigate drug utilization, we also construct a dataset at the patient-year level. On average patients aged 65-75 have 1.05 prescription for young diseases and 2.85 unique prescriptions for old diseases.

4 Impact of the Memorandum on Clinical Trials

We investigate how the Medicare Memorandum of 2000, which lowered financial barriers to participant enrollment, affected clinical trials. We begin by documenting disparities in clinical research before and after the Medicare Memorandum. We then explore how the memorandum shaped the number of clinical trials initiated in high Medicare share diseases. We also consider how the memorandum shaped clinical trial enrollment criteria and the inclusion of elderly participants in those trials.

4.1 Disparities in Clinical Research

We begin by describing how elderly representation in clinical trial enrollment compares to the disease burden. For each ICD-9 code, we calculate both the share of elderly participants in clinical trials and the share of the disease burden that affects the elderly.

Figure 1 compares trial enrollment and disease burden at the ICD-9-level. The 45-degree line represents perfect parity between disease burden and trial enrollment. From 1995-1999, all diseases fell below the 45-degree line, which corresponds to elderly underrepresentation in trials. However, this pattern changed after 2000: the estimated slope between elderly disease burden and trial

enrollment increased from 0.065 to 0.467, which represents greater parity. The difference in slope between 1995-1999 and 2001-2005 is statistically significant, showing that age-based disparities in trial enrollment were reduced after the memorandum.

4.2 Empirical Strategy

We examine the memorandum’s effect on clinical trials using a DID design. Our baseline specification compares outcomes in ICD-9 codes with an above median share of Medicare recipients versus a below median share of Medicare recipients, before and after the Medicare Memorandum of 2000:

$$y_{jt} = \sum_t \beta_t \mathbf{1}\{quarter = t\} \times HighMcareShare_j + HighMcareShare_j + \gamma_{jt} \delta_{c(j)} + \delta_j + \delta_t + \epsilon_{jt} \quad (1)$$

where j is ICD-9 disease code and t refers to quarters of a calendar year. The outcome y_{jt} presents the log number of trial counts within a disease and quarter. We log outcomes since most of our outcomes have a skewed distribution. The coefficient $HighMcareShare_j$ is an indicator for whether the ICD-9 code had an above median share of Medicare patients before 2000. The regression also includes controls for whether the ICD-9 code had an above median share of Medicare patients, ICD-9 fixed effects (δ_j), and quarter fixed effects (δ_t). To account for time-varying differences across disease markets, we include interactions of linear quarter with the ICD-9 disease chapter, which is a broader ICD-9 categorization ($\delta_{c(j)}$) such as "neoplasms." Standard errors are clustered at the ICD-9 level.

In our event studies, we set the reference period as the two years prior to the policy, and constrain the *average* of these coefficients to be zero. Relative to omitting a single time period prior to treatment, using a wider reference period yields a uniform level-shift of the coefficients. This approach helps improve precision, as a single reference point may be noisily estimated, but does not change the relative time patterns or the interpretation of any pre-trends (Miller, 2023). Appendix Figure A.5 shows patterns are similar using the standard choice to omit just the time period prior to treatment.

Descriptive evidence provides support for our empirical strategy. Appendix Figure A.6 plots

the total number of trials within the top 10 percent, 10-20 percent, etc. of diseases by share of Medicare patients. The number of trials per disease category is roughly parallel before 2000, but diseases with high shares of old diagnoses have more trials afterward.

4.3 Impact on the Rate of Clinical Trials

To understand how the memorandum shaped research, we begin by exploring whether the memorandum changed the level of research in clinical trials in diseases common among the elderly. We estimate Equation (1), where the outcome is the $\log(1 + \text{total number of clinical trials})$ in a disease and quarter.¹¹ Figure 2 shows that the difference in the log number of clinical trials between diseases with high versus low Medicare shares prior to 2000 is statistically indistinguishable from zero. We observe similar findings when directly examining the total number of clinical trials (see Appendix Figure A.7). After the memorandum, we observe an immediate increase in the number of clinical trials in diseases with a high versus low Medicare share. This increase remains large, positive, and statistically significant through 2005.¹²

We show corresponding estimates in Panel A of Table 1. We include an indicator for after the memorandum ("Post") instead of relative time fixed effects. For ease of interpretation, we show estimates at the disease-year level. The estimates in column 1 reveal that the memorandum is associated with a 24 percent ($e^{0.213}$) relative increase in the number of clinical trials in diseases with high Medicare share in each year after the memorandum.

The relatively quick increase is consistent with the idea that firms were testing drugs that were previously developed drugs and taken "off the shelf." Support for this intuition is provided by Appendix Table A.3, which shows that the increase in total Phase II and III trials (corresponding to Column 1 in Panel A of Table 1) is largely driven by trials testing drugs that had previously

¹¹We concentrate on linear models primarily for three reasons. First, numerous ICD-9-quarter pairs show no clinical trials at all. Second, the assumption required by Poisson models—that the mean and variance of the data are equal—is not met by our data, making this model less suitable. Third, non-linear models cannot accommodate the rich set of controls we use, such as interactions between ICD-9 disease chapters and quarters. However, as a robustness check of our results, we also include an analysis assessing the effect of the memorandum on the likelihood of *any* clinical trial activity and find similar results (see Appendix Table A.2).

¹²We stop our main analysis in 2005 to avoid issues with the introduction of Medicare Part D in 2006. Further, in 2005, several universities settled federal lawsuits regarding inaccurate Medicare billing for clinical trials. Consequently, trial sponsors were increasingly cautious about Medicare billing.

undergone Phase II or III testing, rather than drugs that had never been tested in these phases before. We also find immediate increases for Phase II, III, and IV trials in old diseases, and a delay in the increase for Phase I trials, which is consistent with an "off the shelf" model (see Appendix Figure A.8).

The effects of the memorandum are likely to be greater in diseases where the cost of participant enrollment is historically high. Appendix Table A.4 shows that the rate of clinical trials disproportionately increases among diseases with longer trials and with inpatient trials, both of which are associated with higher costs (though the difference is not statistically significant, likely due to the small sample size). These results are consistent with the view that the memorandum effectively lowered trial costs in diseases that were more likely to face challenges in participant enrollment.

We evaluate the robustness of our results. For brevity, we briefly outline our main findings and leave a detailed discussion for Appendix A.1. First, we examine the role of concurrent shocks. Figure 2 shows that the increase in trials occurs before Medicare Part D was signed or implemented. Additionally, our results hold after restricting the sample to before 2003 (Appendix Table A.5) and after directly controlling for the passage of Part D (Appendix Table A.6). Our results are robust to using a continuous measure of exposure to the memorandum (Appendix Figure A.2) and to excluding diseases likely to be affected FDA Modernization Act of 1997 (Table A.7). In addition, we show that the memorandum had muted effects in Europe (Appendix Figure A.3), suggesting that our results are not driven by a global shift toward elderly drugs. Finally, we investigate the concern that projections about the United States elderly population shifted around 2000.

4.4 Impact on Clinical Trial Enrollment

4.4.1 Enrollment Criteria

Panel A of Figure 3 examines the impact of the memorandum on drug manufacturers' enrollment criteria for clinical trials. Each clinical trial has specific inclusion and exclusion criteria which include age, diagnoses, health, and prior treatment. We categorize trials as prioritizing elderly participants if the inclusion criteria include *any* adults at age 65 or over, encompassing trials with only elderly participants or both elderly and non-elderly participants. We compare the impact of

the memorandum on these trials against those whose inclusion criteria include *only* non-elderly participants. Panel A of Figure 3 shows that increased exposure to the memorandum is associated with an immediate and sustained increase in the number of trials enrolling any elderly participants. We find no change in the number of trials enrolling non-elderly individuals.

Columns 2 and 3 of Table 1 present seemingly unrelated regressions, which permit a direct comparison of coefficients. Among affected diseases, there was a 26 percent ($e^{0.234}$) yearly increase in the number of trials recruiting any elderly participants. There was no impact on trials recruiting non-elderly participants. Therefore, the increase in trials in elderly diseases following the memorandum consisted only of trials targeting the enrollment of elderly participants.

4.4.2 Enrollment Counts

We now examine whether the shift in trial recruitment translated to an increase in the *actual* quantity of elderly participants. These results are less precise, since we only observe enrollment information for 22% of all trials and 11% of trials before 2000.

We present summary statistics on participant ages before and after the policy in Appendix Table A.8. The mean age of enrollees in old diseases increased from 51.5 to 55.5 (p-value = 0.03), while the mean age of enrollees in young diseases only increased from 38.5 to 41 (p-value = 0.50). Enrollees over age 65 increased from 7 percent to 20 percent in old diseases (p-value = 0.03), compared to 0 percent to 4 percent in young diseases (p-value = 0.19). The log count of enrollees over 65 increased overall, while the log count of young enrollees was unchanged.

Panel B of Figure 3 illustrates the policy’s effect on the percentage of enrollees over age 65. Before the memorandum, the coefficients are statistically insignificant. Beginning with trials that started in June 2000 after the policy, the share of elderly enrollees increased and remained elevated. In column 4 of Table 1, we find the policy led to a statistically insignificant 14 percentage point increase in elderly enrollment, a 100% increase relative to the mean elderly enrollment of 13 percent.

Taken together, our findings suggest the memorandum lowered elderly enrollment costs. As a result, firms conducted more clinical trials for high Medicare share diseases. Firms expanded trial enrollment criteria to include the elderly and suggestive evidence suggests they were successful at recruiting more elderly individuals, with no change in the number of non-elderly individuals.

5 Impact on Drug Utilization

Increases in representation in the R&D process may influence product adoption and utilization. Clinical trial results often appear in marketing materials which may highlight the representation or efficacy of a drug among elderly participants. Physicians may learn about new research from pharmaceutical representatives, publications, or medical guidelines.

We compare drug utilization among elderly versus middle-aged individuals, before and after the Medicare Memorandum of 2000:

$$y_{it} = \sum_t \beta_t \mathbf{1}\{year = t\} \times Elderly_{a(i)} + Elderly_{a(i)} + \delta_t + \eta_i + \epsilon_{it} \quad (2)$$

where y_{it} is the natural log number of unique prescriptions (plus one) among either old or young diseases. This regression is at the patient i and year t level.¹³ Prescriptions are matched to ICD-9 codes and classified as treating old or young diseases as described in section 4. Age categories are denoted by $a(i)$ and $Elderly_{a(i)}$ refers to whether the individual is aged 65-75, relative to a control group of individuals aged 50-60. The δ_t are calendar year fixed effects and η_i are person-level controls for gender, race and ethnicity. Standard errors are robust and we use MEPS analytical weights.

Figure 4 shows that following the memorandum, we observe a substantial increase in the use of elderly-oriented drugs among elderly patients, compared with both drugs targeted towards younger diseases and drug utilization patterns among non-elderly patients. Before the memorandum, use of drugs for old diseases for elderly patients was slightly increasing relative to younger patients, but was closely aligned with patterns for drugs in young diseases. After the memorandum, the use of drugs for old diseases by the elderly began to increase and the difference between old and young diseases became statistically significant in 2002. Table 1, column 6 shows that elderly individuals used 11 percent ($e^{0.104}$) more unique prescriptions for old diseases, compared to no change in prescriptions for young diseases (column 5).

Elderly drug utilization rose by 2002, before Medicare Part D was signed or enacted. As a robustness check that our results are not driven by anticipatory effects of Medicare Part D, we

¹³In some years, survey respondents are asked what date they began their medication, but most dates reported are January 1st of the survey year.

restrict our sample to individuals who earn above 400% of the federal poverty line and have private health insurance (columns 7 and 8). In 1999, 86% of this group already had drug insurance.¹⁴ Among this group, elderly patients reported an 11 percent ($e^{0.100}$) increase in unique prescriptions for old diseases, closely aligning with our main estimates.

Previous studies have found that the implementation of Medicare Part D in 2006 also resulted in an increase in drug utilization. [Duggan and Morton \(2010\)](#) find that drugs most often used by Medicare recipients increased in utilization by 28 percent to 70 percent from 2003 to 2006, though most of their estimates are not statistically significant. [Zhang et al. \(2009\)](#) and [Lichtenberg and Sun \(2007\)](#) also find increases in utilization, but their data begin in 2004 and do not examine the early 2000s. We extend our utilization analysis in levels from 1995 through 2010 in Appendix Figure A.9 and compare the Medicare Memorandum’s impact with Part D. While prescriptions rose modestly after 2003 and 2006, prescription patterns for elderly individuals taking drugs for old diseases had already increased by 2003. These patterns were missed by prior studies that started in 2003 or 2004. Appendix Figure A.9 also shows that the prescription increase in 2006 was mostly driven by drugs for *young* diseases, unlike the increase we find after the memorandum.

6 Why Would Representation Affect Utilization?

Increased representation could shape drug utilization by improving real or perceived drug quality. While quality is difficult to measure, we examine two proxies: post-market adverse events, which reflect a drug’s safety and effectiveness for the target population, and the precision of trial results.

First, we would expect the memorandum—which increased the number of elderly participants enrolled in each trial—would increase the precision of clinical trial results. Our data (which spans 1995-2005) covers only a subset of trials with reported results as mandatory results reporting to ClinicalTrials.gov began in 2007. Appendix Table A.9 shows that the memorandum led to more trials producing precise estimates. In particular, the number of trials with statistically significant outcomes ($p < 0.05$) increased more than those without statistically significant outcomes ($p \geq 0.05$), although the differences between these two measures are not statistically different.

Second, we examine adverse events. Appendix Section A.2 describes the construction of the

¹⁴We cannot directly restrict to individuals with drug insurance as this question was not asked from 1996-1998.

adverse events data. Appendix Table A.9, column 3, shows that on average the elderly experienced 0.16 fewer adverse events per 1,000 prescriptions in old diseases after 2000, a 10 percent decrease from a mean adverse event rate of 1.6. The adverse event rate in other age and disease groups showed no change or even increased.

Adverse events data are subject to reporting bias and requires estimating the total denominator of prescription users in MEPS data which introduces additional noise. However, these results are consistent with the view that increased representation in R&D process may have improved both trial precision and prescription quality.

7 Conclusion

Amid rising public pressure, some policy makers and researchers are exploring strategies to reduce demographic disparities in the innovation process. Others argue that representation in research does not affect innovation or real-world outcomes. We investigate how insurance policies influence representation in drug development. Diseases more affected by the Medicare Memorandum see a 24 percent increase in clinical trials. Within these diseases, firms expanded enrollment criteria to include elderly participants, and we find suggestive evidence that enrollment of elderly individuals increased. Drug utilization among the elderly also rose significantly, suggesting that Medicare-driven representation shifts can meaningfully influence drug use patterns.

Our results offer insights for future research. Representation may affect drug utilization through perceived or actual quality improvements. We provide suggestive evidence that increased representation improves trial precision and quality. However, data limitations prevent a full analysis of how trial outcomes, such as efficacy or survival improvements, change with representation.

Greater representation may also delay trials. The policy we studied subsidized elderly representation, which benefited pharmaceutical firms. Other potential policies could require representation without increasing funding, which may impose costs on firms and delay product approvals.

Our results on age-concordance offer insights into how policies might improve other demographic disparities. For example, in 2022, Medicaid began to cover routine costs for clinical trials, potentially shifting the representation of low income individuals (Takvorian, Guerra and Schpero, 2021). Our results suggest that well-designed policies can address persistent under-representation in innovation.

References

- Aaron, Henry J, and Hellen Gelbrand.** 2000. “Extending Medicare reimbursement in clinical trials.” *National Academies Press*.
- Acemoglu, Daron, David Cutler, Amy Finkelstein, and Joshua Linn.** 2006. “Did medicare induce pharmaceutical innovation?” *American Economic Review*, 96(2): 103–107.
- Alpert, Abby.** 2016. “The anticipatory effects of Medicare Part D on drug utilization.” *Journal of Health Economics*, 49: 28–45.
- Alsan, Marcella, and Marianne Wanamaker.** 2017. “Tuskegee and the Health of Black Men*.” *The Quarterly Journal of Economics*, 133(1): 407–455.
- Alsan, Marcella, Maya Durvasula, Harsh Gupta, Joshua Schwartzstein, and Heidi Williams.** 2024. “Representation and Extrapolation: Evidence from Clinical Trials.” *Quarterly Journal of Economics*, 139(1): 575–635.
- Alsan, Marcella, Romaine A. Campbell, Lukas Leister, and Ayotomiwa Ojo.** 2025. “Investigator racial diversity and clinical trial participation.” *Journal of Health Economics*, 100: 102968.
- Blume-Kohout, Margaret E, and Neeraj Sood.** 2013. “Market size and innovation: Effects of Medicare Part D on pharmaceutical research and development.” *Journal of Public Economics*, 97: 327–336.
- Cao, Ruiqing, Rembrand Koning, and Ramana Nanda.** 2023. “Sampling Bias in Entrepreneurial Experiments.” *Management Science*.
- Cortellis Clinical Trials Intelligence Database.** 1995-2005. *Clarivate Analytics*. <https://clarivate.com/products/cortellis-family/>.
- Cox, KM, S Goel, RL O’connell, M Boyer, PJ Beale, RJ Simes, and MR Stockler.** 2011. “Randomized cross-over trial comparing inpatient and outpatient administration of high-dose cisplatin.” *Internal Medicine Journal*, 41(2): 172–178.

- DiMasi, Joseph A, Henry G Grabowski, and Ronald W Hansen.** 2016. “Innovation in the pharmaceutical industry: new estimates of R&D costs.” *Journal of Health Economics*, 47: 20–33.
- DiMasi, Joseph A, Ronald W Hansen, and Henry G Grabowski.** 2003. “The price of innovation: new estimates of drug development costs.” *Journal of Health Economics*, 22(2): 151–185.
- Duggan, Mark, and Fiona Scott Morton.** 2010. “The effect of Medicare Part D on pharmaceutical prices and utilization.” *American Economic Review*, 100(1): 590–607.
- FDA Adverse Events Reporting System.** 1969-2006. Obtained via FOIA request through Investigative Reporters and Editors <https://www.ire.org/>.
- Friedrichs, Anke, Maren Spies, Martin Härter, and Angela Buchholz.** 2016. “Patient preferences and shared decision making in the treatment of substance use disorders: a systematic review of the literature.” *PloS one*, 11(1): e0145817.
- Global Burden of Disease.** 2019. <https://www.healthdata.org/research-analysis/gbd-data>.
- Green, Angela K., Niti Trivedi, Jennifer J. Hsu, Nancy L. Yu, Peter B. Bach, and Susan Chimonas.** 2022. “Despite The FDA’s Five-Year Plan, Black Patients Remain Inadequately Represented In Clinical Trials For Drugs.” *Health Affairs*, 41(3): 368–374.
- Gupta, Harsh.** 2022. “Do Female Researchers Increase Female Enrollment in Clinical Trials?” *Working Paper*.
- Hebert, Camille.** 2019. “Gender Stereotypes and Entrepreneur Financing.” *Working Paper*.
- Herrera, Angelica P., Shedra Amy Snipes, Denae W. King, Isabel Torres-Vigil, Daniel S. Goldberg, and Armin D. Weinberg.** 2010. “Disparate Inclusion of Older Adults in Clinical Trials: Priorities and Opportunities for Policy and Practice Change.” *American Journal of Public Health*, 100(S1): S105–S112.

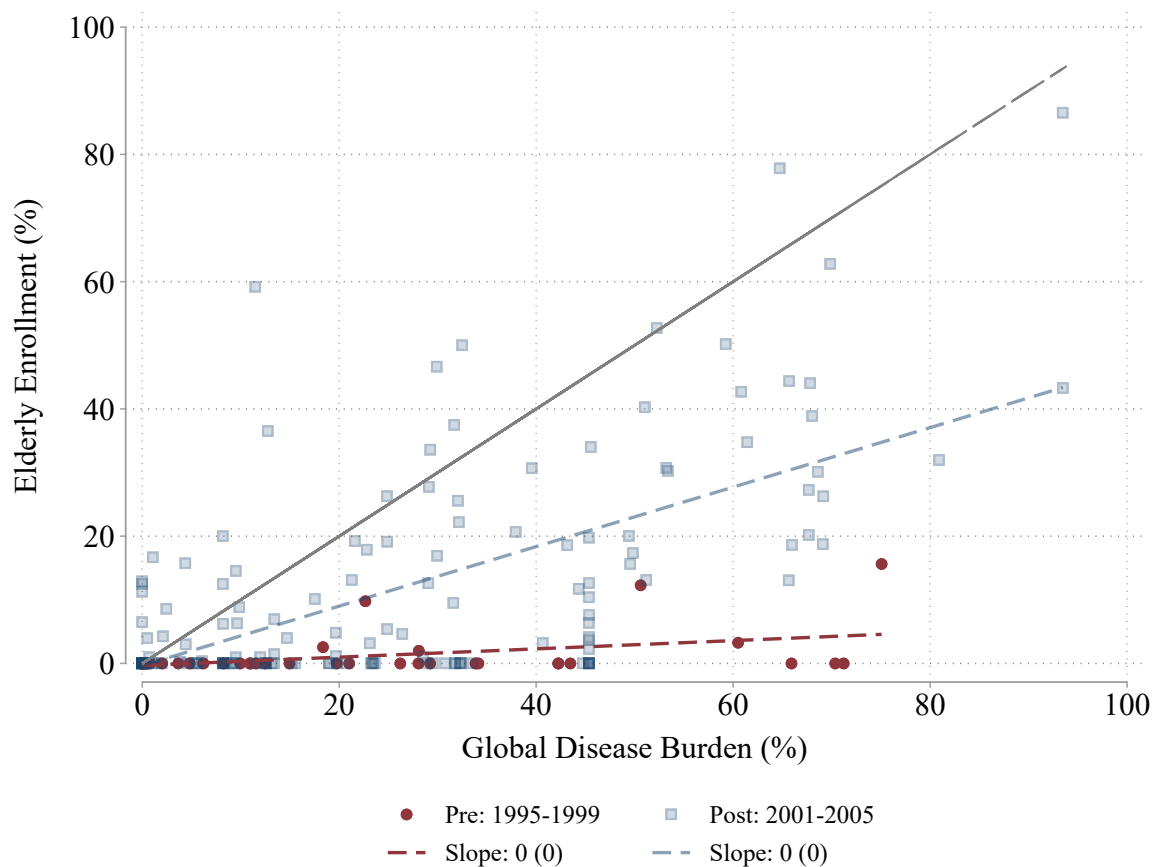
- Hutchins, Laura F, Joseph M Unger, John J Crowley, Charles A Coltman Jr, and Kathy S Albain.** 1999. “Underrepresentation of patients 65 years of age or older in cancer-treatment trials.” *New England Journal of Medicine*, 341(27): 2061–2067.
- Koning, Rembrand, Sampsa Samila, and John-Paul Ferguson.** 2021. “Who do we invent for? Patents by women focus more on women’s health, but few women get to invent.” *Science*, 372(6548): 1345–1348.
- Lara, Primo N., Roger Higdon, Nelson Lim, Karen Kwan, Michael Tanaka, Derick H.M. Lau, Ted Wun, Jeanna Welborn, Frederick J. Meyers, Scott Christensen, Robert O’Donnell, Carol Richman, Sidney A. Scudder, Joseph Tuscano, David R. Gandara, and Kit S. Lam.** 2001. “Prospective Evaluation of Cancer Clinical Trial Accrual Patterns: Identifying Potential Barriers to Enrollment.” *Journal of Clinical Oncology*, 19: 1728–1733.
- Lau, SW Johnny, Yue Huang, Julie Hsieh, Shenggang Wang, Qi Liu, Patricia W Slatum, Janice B Schwartz, Shiew-Mei Huang, and Robert Temple.** 2022. “Participation of older adults in clinical trials for new drug applications and biologics license applications from 2010 through 2019.” *JAMA Network Open*, 5(10): e2236149–e2236149.
- Lichtenberg, Frank R., and Shawn X. Sun.** 2007. “The Impact Of Medicare Part D On Prescription Drug Use By The Elderly.” *Health Affairs*, 26(6).
- McKibbin, Rebecca.** 2023. “The effect of RCTs on drug demand: Evidence from off-label cancer drugs.” *Journal of Health Economics*, 90.
- Medical Expenditure Panel Survey.** 1996-2010. *Agency for Healthcare Research and Quality*. https://meps.ahrq.gov/mepsweb/data_stats/data_overview.jsp.
- Michelman, Valerie, and Lucy Msall.** 2022. “Sex, Drugs, and R&D: Missing Innovation from Regulating Female Enrollment in Clinical Trials.” *Working Paper*.
- Miller, Douglas L.** 2023. “An Introductory Guide to Event Study Models.” *Journal of Economic Perspectives*, 37(2): 203–30.

- Nadel, Mark V.** 1992. “Women’s Health: FDA Needs to Ensure More Study of Gender Differences in Prescription Drug Testing.” *Washington, DC: US House of Representatives General Accounting Office*.
- Nielsen, Mathias Wullum, Jens Peter Anderson, Londa Schiebinger, and Jesper W. Schneider.** 2017. “One and a half million medical papers reveal a link between author gender and attention to gender and sex analysis.” *Nature Human Behavior*, 1: 791–796.
- Pear, Robert.** 2000. “Clinton to Order Medicare to Pay New Costs.” *The New York Times*, A p.24.
- Ristovska, Ljubica.** 2025. “Drug Adoption under Uncertain Quality and Impact of FDA (De)Certification.” *Working Paper*.
- Sertkaya, Aylin, Anna Birkenbach, Ayesha Berlind, and John Eyraud.** 2014. “Examination of Clinical Trial Costs and Barriers for Drug Development.” *U.S. Department of Health and Human Services*.
- Sertkaya, Aylin, Hui-Hsing Wong, Amber Jessup, and Trinidad Beleche.** 2016. “Key cost drivers of pharmaceutical clinical trials in the United States.” *Clinical Trials*, 13(2): 117–126.
- Takvorian, Samuel U., Carmen E. Guerra, and William L. Schpero.** 2021. “A Hidden Opportunity — Medicaid’s Role in Supporting Equitable Access to Clinical Trials.” *New England Journal of Medicine*, 384(21): 1975–1978.
- Unger, Joseph M, Charles A Coltman Jr, John J Crowley, Laura F Hutchins, Silvana Martino, Robert B Livingston, John S Macdonald, Charles D Blanke, David R Gandara, E David Crawford, et al.** 2006. “Impact of the year 2000 Medicare policy change on older patient enrollment to cancer clinical trials.” *Journal of Clinical Oncology*, 24(1): 141–144.
- U.S. Census Bureau.** 2023. “Census Bureau Releases New 2020 Census Data on Age, Sex, Race, Hispanic Origin, Households and Housing.” <https://www.census.gov/newsroom/press-releases/2023/2020-census-demographic-profile-and-dhc.html>.

Zhang, Yuting, Julie M. Donohue, Judith R. Lave, Gerald O'Donnell, and Joseph P. Newhouse. 2009. "The Effect of Medicare Part D on Drug and Medical Spending." *New England Journal of Medicine*, 361(1): 52–61.

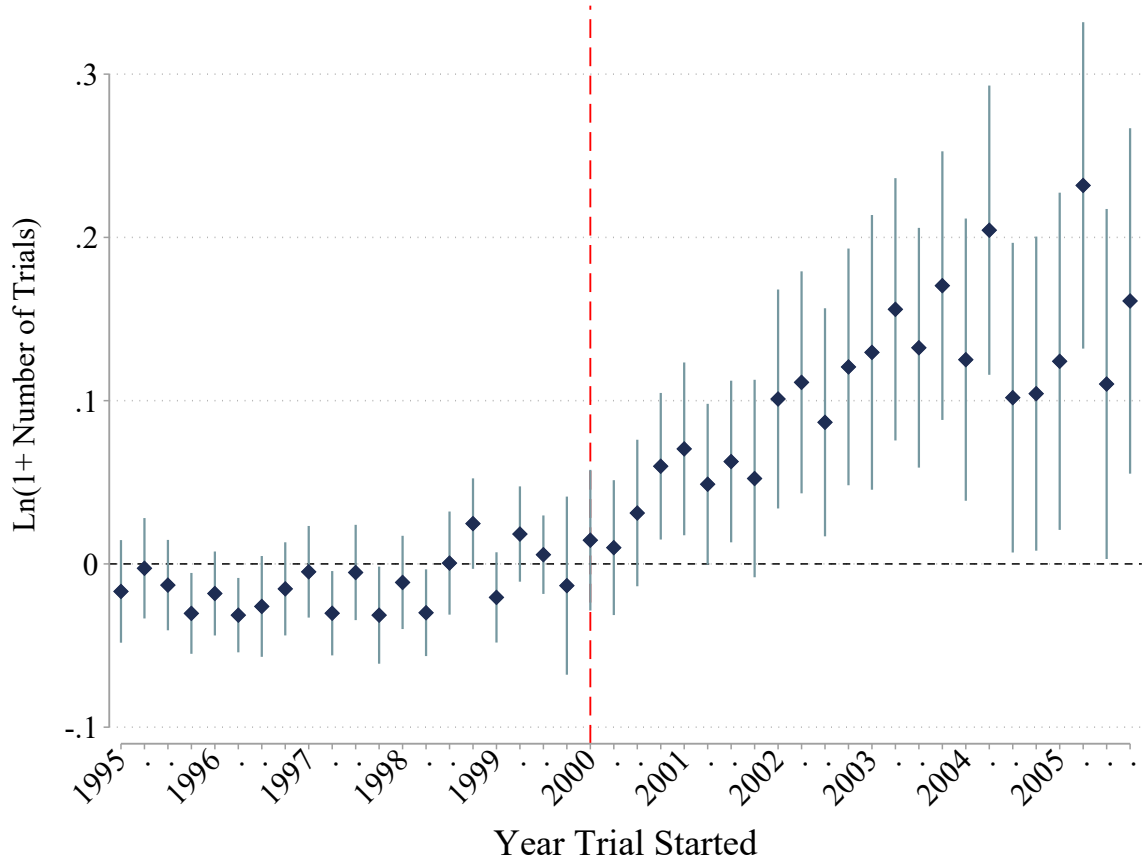
Figures and Tables

Figure 1: DISPARITIES IN ELDERLY ENROLLMENT OVER TIME



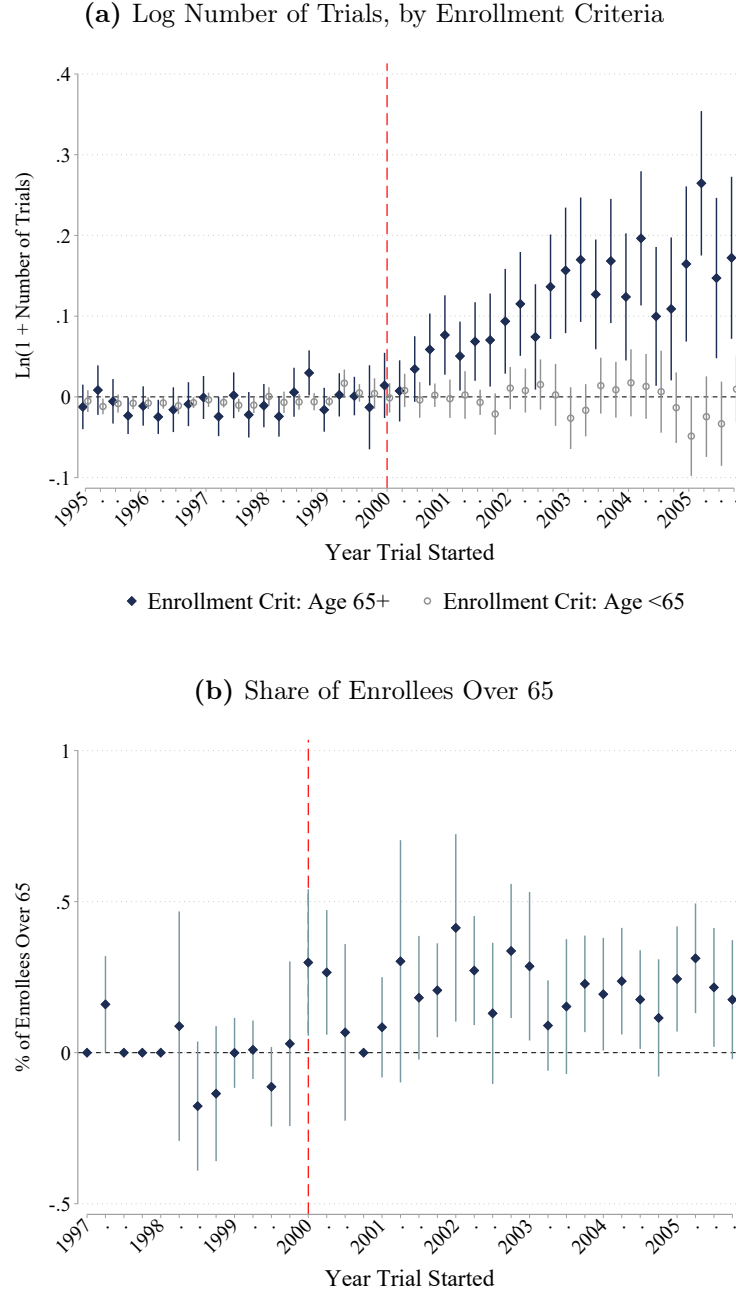
NOTE: Each point represents an ICD-9 code, with the x-axis showing the share of the disease burden that is among the elderly (aged over 65) and the y-axis presenting the average share of elderly participants in clinical trials for those diseases in our sample. The 45-degree line represents perfect parity between disease burden and trial enrollment. Points below this line indicate diseases where the elderly disease burden exceeds elderly trial enrollment. The darker dots average across trials that begin from 1995-1999 while the lighter squares average across 2001-2005. These two lines have statistically different slopes with $p < 0.001$.

Figure 2: IMPACT OF MEDICARE REIMBURSEMENT EXPANSION ON TRIALS



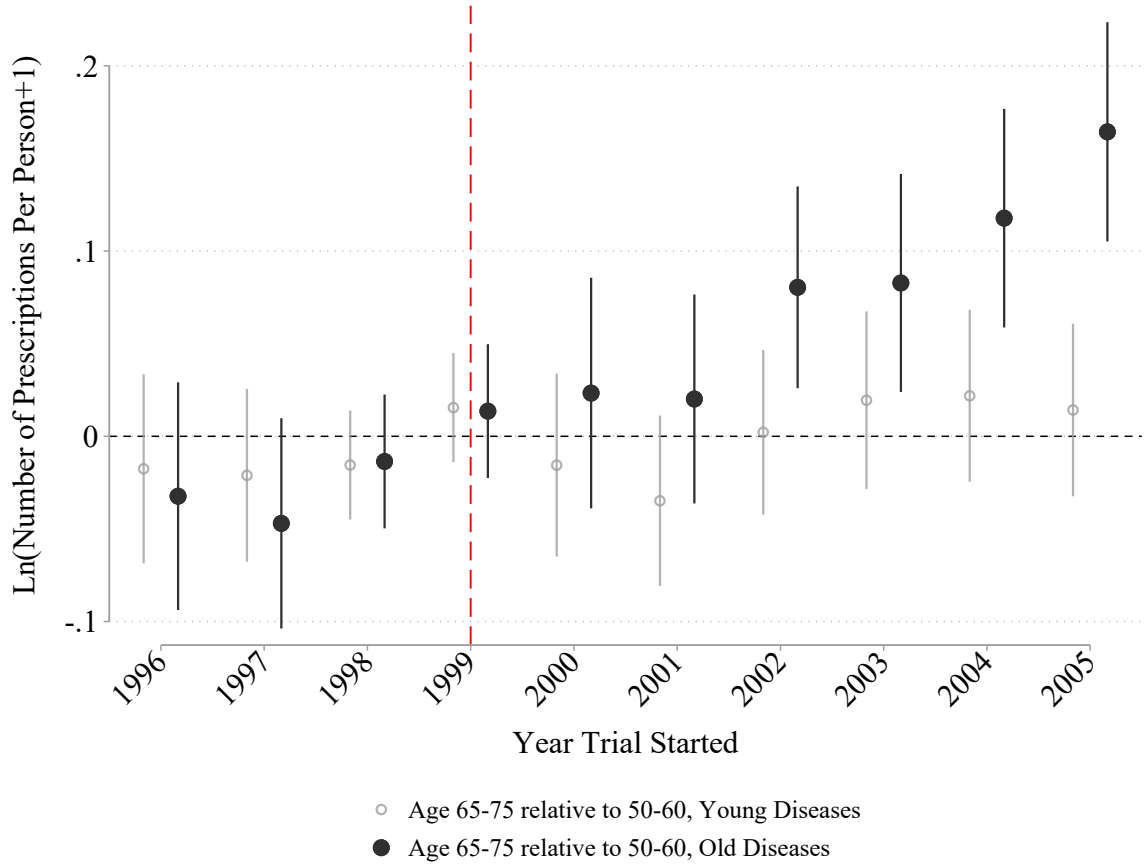
NOTE: Figure presents event study coefficients from the estimation of equation 1. Observations are at the disease-quarter level. We plot the coefficients on the interaction between quarter and an indicator for whether a disease had an above median share of diagnoses among the elderly from 1996-1999. The outcome is the natural log of the number of trials in a quarter plus one. The dotted vertical line represents the quarter before the memorandum.

Figure 3: IMPACT OF MEDICARE REIMBURSEMENT EXPANSION ON ENROLLMENT



NOTE: Figure presents event study coefficients from the estimation of equation 1. Observations are at the disease-quarter level. We plot the coefficients on the interaction between quarter and an indicator for whether a disease had an above median share of diagnoses among the elderly from 1996-1999. Panel A presents estimates from two different regressions. In one the outcome is the natural log of the number of trials that include elderly participants plus one. In the other the outcome is the natural log of the number of trials that do not include elderly participants in their enrollment criteria plus one. All trials are in one of these two categories. In Panel B the outcome is the share of enrollees that are older than 65. There are only three trials with enrollment information before 1997, so we drop years before 1997. Disease-year combinations with no enrollment information are considered missing. The dotted vertical line represents the quarter before the memorandum.

Figure 4: LOG PRESCRIPTIONS BY AGE AND DISEASE CATEGORY



NOTE: Figure presents event study coefficients from the estimation of equation 2. Observations are at the patient-year level. We plot the coefficients on the interaction between an indicator for age 65-75 (relative to 50-60) and year. The outcome is the natural log of the number of unique prescriptions per person (plus one) in diseases with an above median share of diagnoses among the elderly ("Old diseases"), or in diseases with a below median share of diagnoses among the elderly ("Young Diseases"). The dotted vertical line represents the year before the memorandum.

Table 1: IMPACT ON CLINICAL TRIALS AND PRESCRIPTIONS*A. Disease-Year Level Analysis, Impact on Trials*

	Ln(1 + # Total Trials) (1)	Ln(1 + # Trials Enrolling Only Age < 65) (2)	Ln(1 + # Trials Enrolling Any Age ≥ 65) (3)	Share of Participants Age ≥ 65 (4)
Post × MedicareShare	0.213*** (0.0538)	0.00168 (0.0233)	0.234*** (0.0501)	0.136 (0.0987)
Post	-0.168*** (0.0311)	-0.0252 (0.0166)	-0.175*** (0.0287)	0.0364 (0.0834)
Mean of Dep. Var.	0.336	0.079	0.298	0.129
Diff. Wald Test P-value, Col. (2) vs. (3)		0.00		
Observations	4,290	4,290	4,290	478
Adjusted R^2	0.602	0.233	0.603	0.312
ICD9 FEs	Yes	Yes	Yes	Yes
ICD9 Category × Year Trends	Yes	Yes	Yes	Yes

B. Patient-Level Analysis, Impact on Prescriptions

	Full Sample		High Income, Private Insurance	
	Ln(1 + # Prescriptions)		Ln(1 + # Prescriptions)	
	Young Diseases (5)	Old Diseases (6)	Young Diseases (7)	Old Diseases (8)
Post × Elderly	0.0126 (0.0128)	0.104*** (0.0158)	-0.00676 (0.0222)	0.100*** (0.0269)
Post	0.0232* (0.00913)	0.0668*** (0.0109)	0.0186 (0.0129)	0.0708*** (0.0152)
Mean of Dep. Var.	0.497	0.830	0.508	0.738
Diff. Wald Test p-value, Col. (1) vs. (2)	0.00		0.00	
Observations	53,539	53,539	19,949	19,949
Adjusted R^2	.068	.084	.081	.08
Race FEs	Yes	Yes	Yes	Yes
Ethnicity FEs	Yes	Yes	Yes	Yes
Gender FEs	Yes	Yes	Yes	Yes

NOTE: Panel A presents coefficients from a disease-year level regression that regresses the outcome in each column on an indicator for after the Medicare Memorandum of 2000 ("Post"), an indicator for whether a disease has an above median share of diagnoses among the elderly ("Medicare Share"), and the interaction between the two. We control for ICD-9 fixed effects (which encompass the "Medicare Share" indicator) and the ICD-9 category by year trends. The "Wald Test P-value" is the p-value comparing the $Post \times MedicareShare$ coefficient in column 2 vs. 3. In Panel A, columns 1 and 4 are estimated using OLS; columns 2 and 3 estimated using seemingly unrelated regressions. Panel B presents coefficients from an patient-level regression that regresses the outcome in each column on an indicator for after the Medicare Memorandum of 2000 ("Post"), an indicator for whether the respondents are over 65 ("Elderly"), and the interaction between the two. We control for an indicator for elderly and person-level controls for race, ethnicity, and gender. In Panel B, columns 5-8 are estimated using seemingly unrelated regressions. High Income refers to individuals above 400% of the federal poverty line. Standard errors are reported in parentheses, with * $p < 0.05$; ** $p < 0.01$ and *** $p < 0.001$.

A Supplemental Appendix

A.1 Additional Robustness Checks

We evaluate the robustness of our results in relation to other shocks. Consistent with the view that our results are not driven by the introduction of Medicare Part D, Figure 2 shows that the increase in trials in diseases with above median elderly diagnoses occurs several years before 2003 (the Medicare Modernization Act was signed into law in December 2003). We also conduct two additional robustness tests: first, in column 1 of Appendix Table A.5, we show that our results are robust to restricting the sample years to 1995 to 2003. Second, we show that our results are robust to including an additional control for the passage of Medicare Part D in 2003. Column 1 of Appendix Table A.6 shows that in a horse-race between the memorandum and Part D, the memorandum has a meaningful impact on the number of total clinical trials. Additionally, our results are robust to using an alternative, continuous measure of exposure to the memorandum (see Appendix Figure A.2). A related policy is the 1997 FDA Modernization Act, which established fast-track approval for drugs treating serious conditions, such as HIV and leukemia. As a robustness check, we exclude all ICD9 codes associated with serious conditions (i.e., linked to drugs receiving fast-track approval). The results, shown in Appendix Table A.7, are consistent with our main results.

We show that our results are not driven by a global shift toward elderly drugs. The Medicare memorandum lowered the cost of trials run in the United States, but did not change the costs of trials in Europe. In Appendix Figure A.3, we show the memorandum had muted effects in European countries, compared to the United States. Innovation in other countries may be a complement or substitute to US-based innovation, but we would expect any spillover effects to be smaller than the main effect of the policy.

Finally, we investigate the concern that projections about the United States elderly population shifted around 2000. If demand for pharmaceuticals by elderly patients were projected to increase, this might motivate more innovation in elderly diseases. However, the 2000 census did not show large gains in elderly population. From 1990 to 2000, the growth of the elderly population was 12.0

percent, compared to the total population growth of 13.2 percent.¹ In 2000, the baby boomers, defined as those born 1946-1964, would still be aged 36-54 and projections about their age patterns did not change discretely.

A.2 Adverse Events Construction

We measure adverse events included in the Adverse Event Reporting System (AERS), a database of reported adverse events used to monitor post-marketing safety by the FDA. We obtained historical AERS files from 1969-2006 through a Freedom of Information Act request initiated by the Investigative Reporters and Editors group ([FDA Adverse Events Reporting System, 1969-2006](#)). Adverse events can be reported by health care professionals and consumers such as patients, family members, lawyers, and others. These may be sent to the FDA or to manufacturers, who are required to forward the report to the FDA. AERS reports the drug name, event year, age and gender of the individual who experienced the event.

We first construct the denominator of total prescriptions using the MEPS. We count the number of unique people taking each prescription by drug, age category, year, and gender, using the MEPS weights to scale these prescription counts to the US population. We then count the number of adverse events reported in AERS within each of these drug, age category, year and gender cells. Our main outcome is the adverse event rate per 1,000 unique prescriptions. As in section 5, we separate these outcomes into prescriptions for old and young diseases.

For each drug, age category, year, and gender cell, we regress

$$y_{d,a,g,t} = \beta_t Post_t Elderly_a + Elderly_a + \delta_{j(d)} + \eta_g + \epsilon_{d,a,g,t} \quad (3)$$

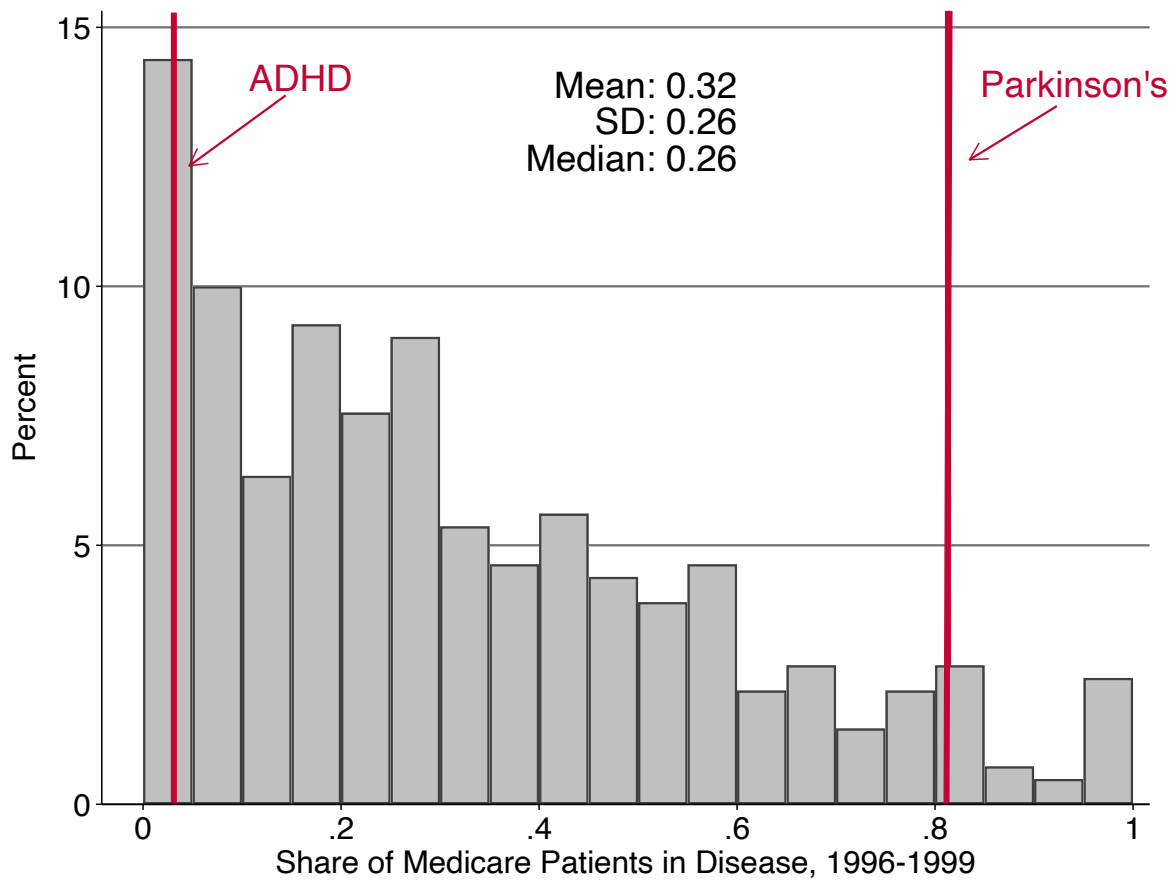
where $y_{d,a,g,t}$ is the adverse event rate for each drug d in age group a and gender g in year t . We run these separately for drugs for either old or young diseases as described in section 4. $Post_t$ is an indicator for after the memorandum. $Elderly_a$ refers to whether the age is 65-75, relative to a control category aged 50-60. $\delta_{j(d)}$ are indicators for the ICD-9 code j for each drug. η_g refer to person-level controls for gender. Standard errors are robust and we include analytical weights of

¹https://www.census.gov/newsroom/releases/archives/2010_census/cb11-cn192.html

the number of MEPS prescriptions in each cell.

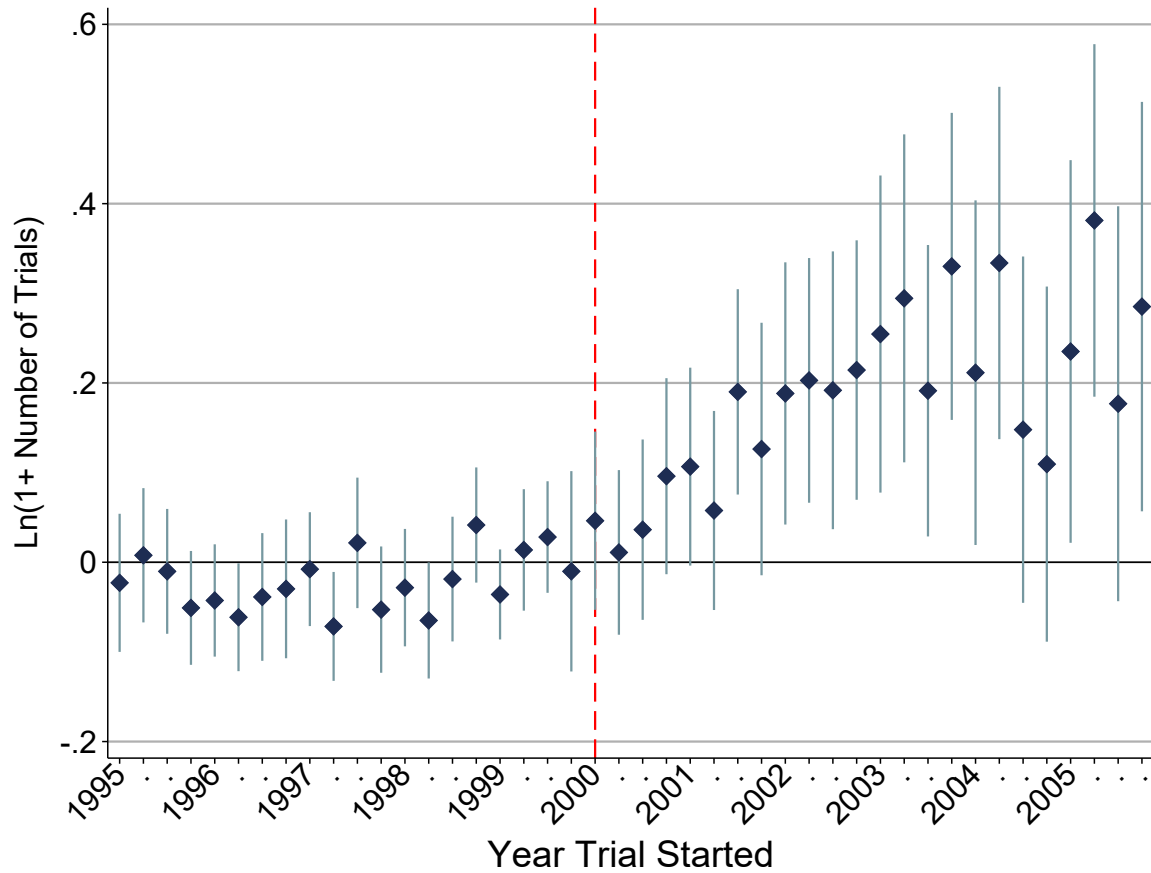
A.3 Additional Tables and Figures

Figure A.1: DISEASE-LEVEL EXPOSURE TO MEDICARE POLICY



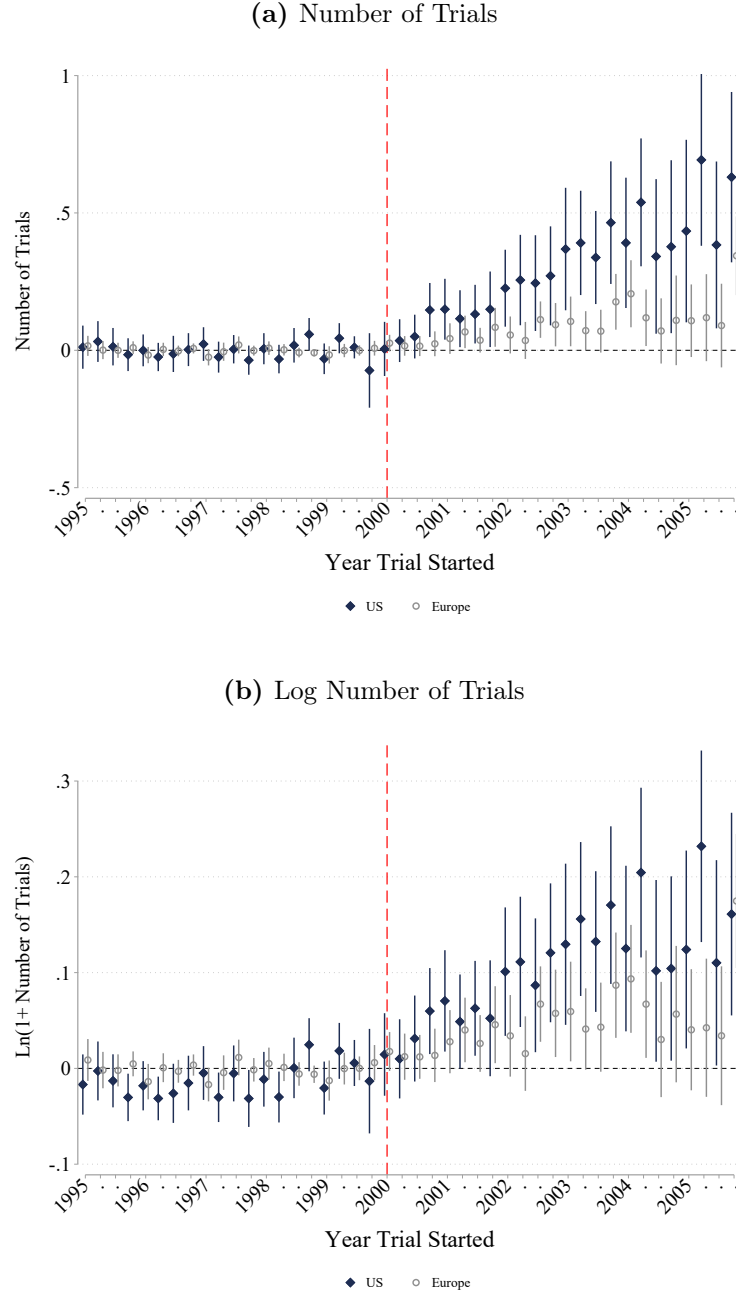
NOTE: Figure plots the share of diagnoses in a disease among Medicare patients. The sample used is the MEPS from 1996-1999.

Figure A.2: IMPACT OF MEDICARE MEMORANDUM USING CONTINUOUS MEDICARE SHARE



NOTE: Figure is similar to figure 2, but uses a continuous rather than binary measure of Medicare share of diagnoses. Observations are at the disease-quarter level and the outcome is the natural log of the number of trials in a disease-quarter. We plot the coefficients on the interaction between year and a continuous measure of the share of diagnoses among Medicare patients from 1996-1999. The dotted vertical line represents the quarter before the memorandum.

Figure A.3: IMPACT OF MEDICARE REIMBURSEMENT IN EUROPE



NOTE: Figure presents event study coefficients from the estimation of equation 1 separately for trials in the United States versus Europe. Observations are at the disease-quarter level and the outcome is the number of trials in a disease-quarter (Panel A) or the natural log of the number of trials in a disease-quarter plus one (Panel B). We plot the coefficients on the interaction between year and an indicator for whether a disease had an above median share of diagnoses among the elderly from 1996-1999. The dotted vertical line represents the quarter before the memorandum.

Figure A.4: CLINICAL TRIAL ENROLLMENT DATA FROM CLINICALTRIALS.GOV

COMPLETED ⓘ

Efficacy and Safety of Adalimumab in Patients With Active Rheumatoid Arthritis Treated Concomitantly With Methotrexate.

ClinicalTrials.gov ID ⓘ NCT00195702

Sponsor ⓘ Abbott

Information provided by ⓘ Abbott (Responsible Party)

AGE, CUSTOMIZED

Measure Type: Number | Unit of measure: participants

Number Analyzed	619 participants
< 40 years	55
Between 40 and 64 years	402
Between 65 and 74 years	130
>= 75 years	32

STUDY START ⓘ

2000-02

PRIMARY COMPLETION (ACTUAL) ⓘ

2002-09

STUDY COMPLETION (ACTUAL) ⓘ

2010-08

ENROLLMENT (ACTUAL) ⓘ

619

STUDY TYPE ⓘ

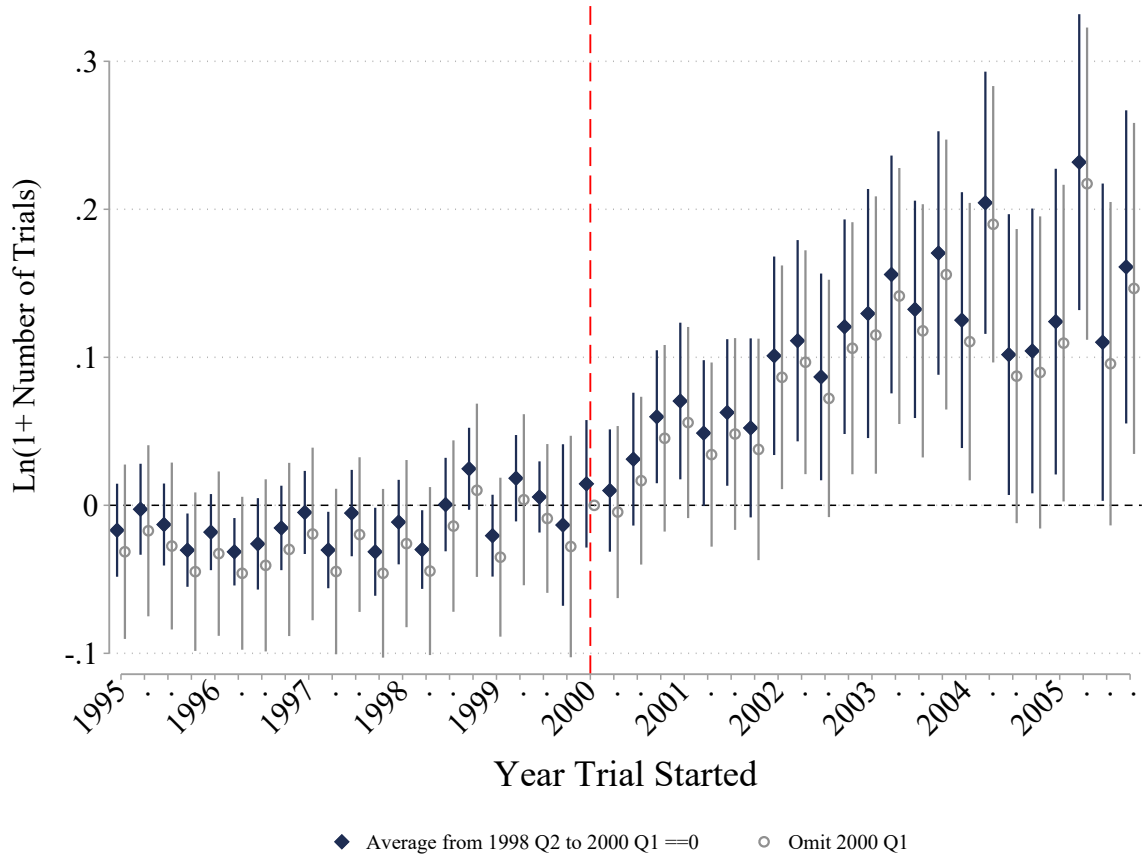
Interventional

PHASE ⓘ

Phase 3

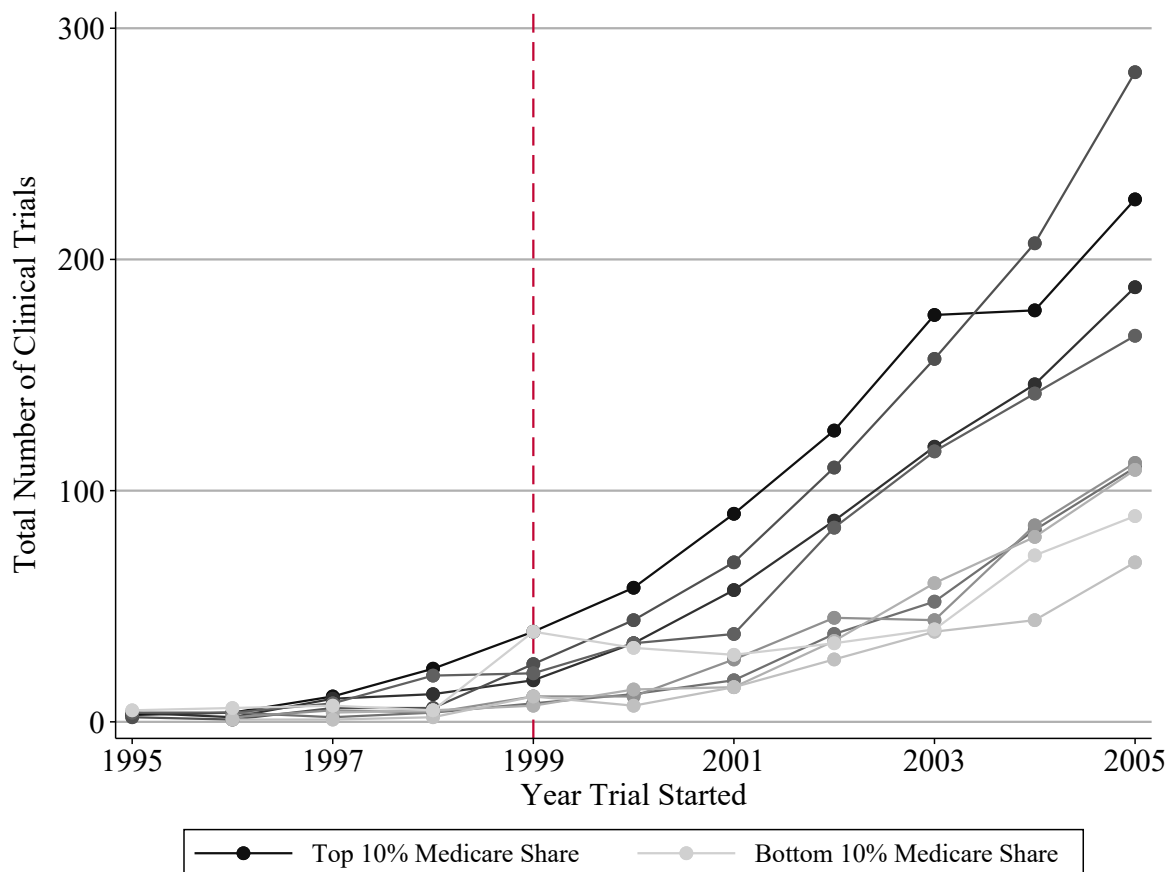
NOTE: Figure shows an example of trial enrollment counts in ClinicalTrials.gov. This trial was a phase 3 trial testing the efficacy of adalimumab. The trial was conducted in 2000, and adalimumab (brand name Humira) was approved by the FDA in 2002.

Figure A.5: CHANGE OMITTED GROUP



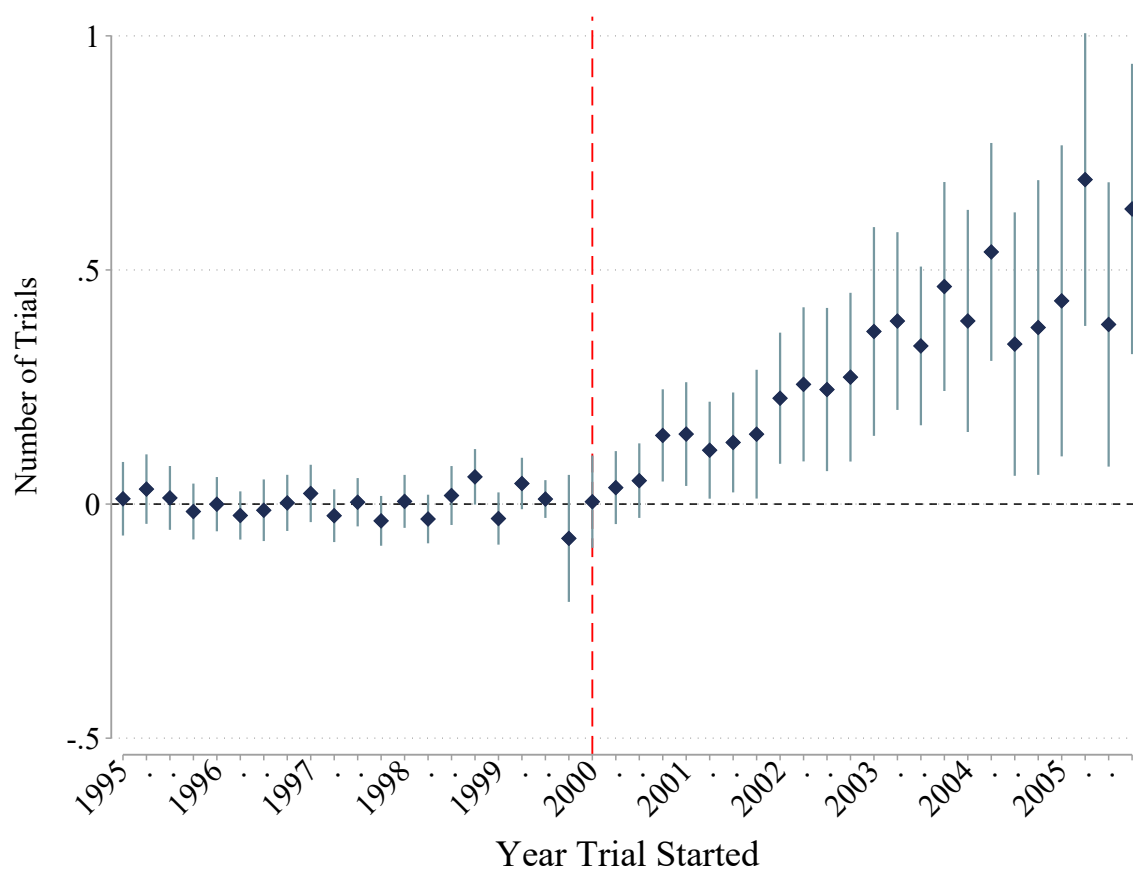
NOTE: Figure is similar to figure 2, but changes the omitted group in the equation estimation. Observations are at the disease-quarter level and the outcome is the natural log of the number of trials in a disease-quarter. The estimates in black diamonds are the same as in the main specification and the estimates in hollow circles omits just the quarter before the policy. The dotted vertical line represents the quarter before the memorandum.

Figure A.6: DESCRIPTIVE EVIDENCE OF MEDICARE POLICY



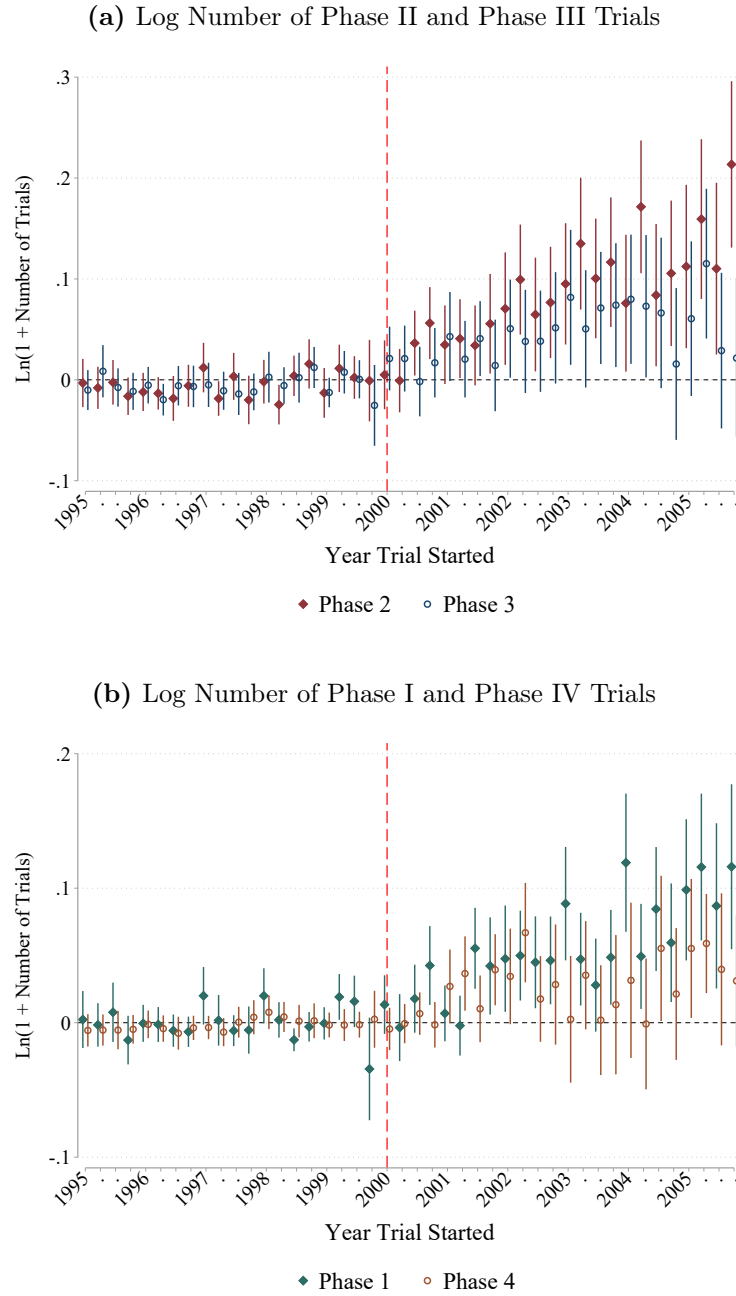
NOTE: Figure plots the total number of clinical trials within disease categories. The darkest line refers to diseases in the top 10 percent of Medicare share of diagnoses. The second-darkest contains the diseases in the top 10-20 percent of Medicare share of diagnoses all the way down to the lightest line which contains the lowest 10 percent of diseases in terms of the Medicare share of diagnoses. The increase in the number of trials for diseases in the bottom 10% Medicare share in 1999 is primarily driven by HIV-related clinical trials. The dotted vertical line represents the year before the memorandum.

Figure A.7: IMPACT OF MEDICARE MEMORANDUM ON NUMBER OF TRIALS



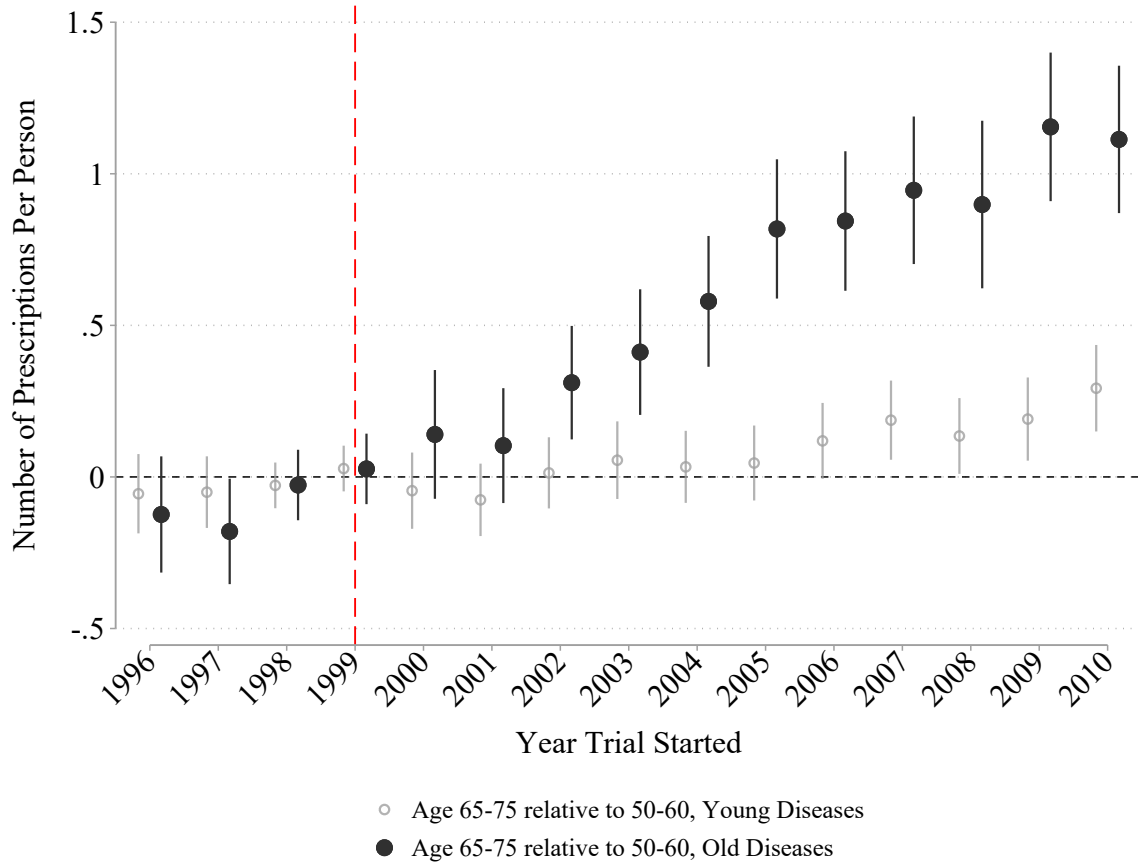
NOTE: Figure is similar to figure 2, but examines the impact on the number of clinical trials. Observations are at the disease-quarter level. We plot the coefficients on the interaction between year and a continuous measure of the share of diagnoses among Medicare patients from 1996-1999. The dotted vertical line represents the quarter before the memorandum.

**Figure A.8: IMPACT OF MEDICARE REIMBURSEMENT EXPANSION ON
NUMBER OF TRIALS, BY TRIAL PHASE**



NOTE: Figure presents event study coefficients from the estimation of equation 1. Observations are at the disease-quarter level. We plot the coefficients on the interaction between quarter and an indicator for whether a disease had an above median share of diagnoses among the elderly from 1996-1999. The outcomes are the natural log of the number of Phase I, Phase II, Phase III, and Phase IV trials. The dotted vertical line represents the quarter before the memorandum.

Figure A.9: PRESCRIPTIONS BY AGE AND DISEASE CATEGORY, EXPANDED YEARS



NOTE: Figure presents event study coefficients from the estimation of equation 2. Observations are at the patient-level. We plot the coefficients on the interaction between an indicator for whether the individual is 65-75 (relative to 50-60) and year. The outcome is the number of unique prescriptions per person in diseases with an above median share of diagnoses among the elderly ("Old diseases"), or a below median share of diagnoses among the elderly ("Young Diseases"). The dotted vertical line represents the year before the memorandum.

Table A.1: SUMMARY STATISTICS

	Count (1)	Mean (2)	Median (3)	SD (4)	Min (5)	Max (6)
<i>A. Disease-Quarter Level</i>						
Medicare Share	4,290	0.32	0.27	0.26	0	1
# Trials: Phase II and III	4,290	1.09	0.00	3.51	0	58
# Trials: Enrolling Only Age < 65	4,290	0.15	0.00	0.63	0	10
# Trials: Enrolling Any Age $\geq$ 65	4,290	0.93	0.00	3.32	0	56
Share of Enrolled Participants Age $\geq$ 65	478	0.13	0.00	0.22	0	1
<i>B. Patient-Year Level</i>						
# Prescriptions						
Patients Age 50-60						
Higher Medicare Share Diseases	34,095	1.67	1.00	2.36	0	25
Low Medicare Share Diseases	34,095	1.04	0.00	1.66	0	24
Patients Age 65-75						
Higher Medicare Share Diseases	19,444	2.85	2.00	2.91	0	24
Low Medicare Share Diseases	19,444	1.05	0.00	1.56	0	20

NOTE: Table presents summary statistics for outcomes at the disease-year level (Panel A), or the patient-year level (Panel B). Medicare share is the share of diagnoses in that disease among Medicare patients. "Criteria Any Age $\geq$ 65" includes the number of trials with enrollment criteria that include elderly individuals. "Criteria Only Age <65" includes the number of trials with enrollment criteria that do not include elderly individuals. Both panels are restricted to years 1995-2005. "# of Prescriptions" refers to the number of unique prescriptions per person in that disease category in the MEPS.

Table A.2: IMPACT ON ANY CLINICAL TRIAL

	1(Any Trial)	1(Any Trial Enrolling Only Age < 65)	1(Any Trial Enrolling Any Age ≥ 65)
	(1)	(2)	(3)
Post × MedicareShare	0.0707* (0.0311)	-0.00142 (0.0212)	0.0882** (0.0303)
Post	-0.0459* (0.0231)	-0.0126 (0.0171)	-0.0671** (0.0221)
Mean of Dep. Var.	0.248	0.086	0.224
Diff. Wald Test p-value			0.00
Observations	4,290	4,290	4,290
Adjusted R^2	0.469	0.267	0.473
ICD9 FEs	Yes	Yes	Yes
ICD9 Category × Year Trends	Yes	Yes	Yes

NOTE: Table presents coefficients from regressing the outcome in each column on an indicator for after the Medicare Memorandum of 2000 ("Post"), an indicator for whether a disease has an above median share of diagnoses among the elderly ("Medicare Share"), and the interaction between the two. Observations are at the disease-year level. We control for ICD-9 fixed effects (which encompass the "Medicare Share" indicator) and the ICD-9 category by year trends. The "Wald Test P-value" is the p-value comparing the $Post \times MedicareShare$ coefficient in columns 2 vs. 3. Column 1 estimated using OLS. Columns 2 and 3 estimated using seemingly unrelated regressions. Standard errors are reported in parentheses, with * $p < 0.05$; ** $p < 0.01$ and *** $p < 0.001$.

Table A.3: IMPACT ON TRIALS TESTING NOVEL DRUGS AND PREVIOUSLY TESTED DRUGS

	Ln(1 + # Trials)	
	Testing Any Novel Drugs (1)	Testing Only Previously Tested Drugs (2)
Post $\times$ MedicareShare	0.125*** (0.0297)	0.185*** (0.0478)
Post	-0.0895*** (0.0190)	-0.158*** (0.0280)
Mean of dep. var.	0.149	0.267
Diff. Wald Test p-value	0.04	
Observations	4290	4290
Adjusted R^2	0.466	0.561
ICD9 FEs	Yes	Yes
ICD9 Category $\times$ Year Trends	Yes	Yes

NOTE: Table presents coefficients from regressing the outcome in each column on an indicator for after the Medicare Memorandum of 2000 ("Post"), an indicator for whether a disease has an above median share of diagnoses among the elderly ("Medicare Share"), and the interaction between the two. Observations are at the ICD-9 code-year level. The outcome in column 1 is the log number (plus 1) of trials that test drugs not previously investigated in any phase II or phase III trial prior to the focal trial. The outcome in column 2 is the log number (plus 1) of trials testing only drugs that have been previously evaluated in prior phase II or phase III trials before the focal trial. We control for ICD-9 fixed effects (which encompass the "Medicare Share" indicator) and the ICD-9 category by year trends. The "Wald Test P-value" is the p-value comparing the $Post \times MedicareShare$ coefficient in columns 1 vs. 2. Estimates come from seemingly unrelated regressions. Standard errors are reported in parentheses, with * $p < 0.05$; ** $p < 0.01$ and *** $p < 0.001$.

Table A.4: HETEROGENEITY ACROSS DISEASES

	Ln(1 + # Trials)			
	Duration		Patient Care Type	
	Short Trials (1)	Long Trials (2)	Outpatient (3)	Inpatient (4)
Post $\times$ MedicareShare	0.141* (0.0617)	0.297*** (0.0882)	0.0814 (0.0545)	0.205* (0.0959)
Post	-0.168*** (0.0355)	-0.177** (0.0552)	-0.123*** (0.0310)	-0.151* (0.0637)
Mean of Dep. Var.	0.253	0.415	0.192	0.487
Diff. Wald Test p-value	0.15		0.26	
Observations	2,332	1,958	2,255	2,035
Adjusted R^2	0.512	0.66	0.468	0.669
ICD9 FEs	Yes	Yes	Yes	Yes
ICD9 Category $\times$ Year Trends	Yes	Yes	Yes	Yes

NOTE: Table presents coefficients from regressing the outcome in each column on an indicator for after the Medicare Memorandum of 2000 ("Post"), an indicator for whether a disease has an above median share of diagnoses among the elderly ("Medicare Share"), and the interaction between the two. Observations are at the disease-year level. We control for ICD-9 fixed effects (which encompass the "Medicare Share" indicator) and the ICD-9 category by year trends. Results are estimated using seemingly unrelated regressions. Standard errors are reported in parentheses, with * $p < 0.05$; ** $p < 0.01$ and *** $p < 0.001$.

Table A.5: IMPACT OF MEDICARE REIMBURSEMENT EXPANSION (1995 TO 2003)

	Ln(1 + # Total Trials) (1)	Ln(1 + # Trials Enrolling Only Age < 65) (2)	Ln(1 + # Trials Enrolling Any Age ≥ 65) (3)	Share of Participants Age ≥ 65 (4)
Post × MedicareShare	0.199*** (0.0455)	0.00167 (0.0187)	0.219*** (0.0425)	0.257 (0.139)
Post	-0.0729** (0.0280)	0.0157 (0.0142)	-0.0924*** (0.0261)	0.158 (0.113)
Mean of Dep. Var.	0.230	0.048	0.203	0.112
Diff. Wald Test p-value			0.00	
Observations	3,510	3,510	3,510	234
Adjusted R^2	0.553	0.226	0.550	0.389
ICD9 FEs	Yes	Yes	Yes	Yes
ICD9 Category × Year Trends	Yes	Yes	Yes	Yes

NOTE: Table presents coefficients from regressing the outcome in each column on an indicator for after the Medicare Memorandum of 2000 ("Post"), an indicator for whether a disease has an above median share of diagnoses among the elderly ("Medicare Share"), and the interaction between the two. Observations are at the disease-year level. We control for ICD-9 fixed effects (which encompass the "Medicare Share" indicator) and the ICD-9 category by year trends. The "Wald Test P-value" is the p-value comparing the $Post \times MedicareShare$ coefficient in columns 2 vs. 3. Columns 1 and 4 are estimated using OLS. Columns 2 and 3 estimated using seemingly unrelated regressions. Standard errors are reported in parentheses, with * $p < 0.05$; ** $p < 0.01$ and *** $p < 0.001$.

Table A.6: COMPARISON BETWEEN MEDICARE MEMORANDUM OF 2000 AND
MEDICARE PART D

	Ln(1 + # Total Trials) (1)	Ln(1 + # Trials Enrolling Only Age < 65) (2)	Ln(1 + # Trials Enrolling Any Age ≥ 65) (3)	Share of Participants Age ≥ 65 (4)
Post × MedicareShare	0.196*** (0.0466)	0.00923 (0.0197)	0.209*** (0.0436)	0.136 (0.0991)
Post	-0.0833** (0.0282)	0.00558 (0.0145)	-0.0950*** (0.0262)	0.0327 (0.0911)
PostPartD × MedicareShare	0.0594 (0.0481)	-0.0263 (0.0303)	0.0867 (0.0445)	-0.00421 (0.0462)
PostPartD	0.131*** (0.0341)	0.0881*** (0.0222)	0.0976** (0.0318)	-0.00142 (0.0465)
Mean of Dep. Var.	0.336	0.079	0.298	0.129
Diff. Wald Test p-value			0.00	
Observations	4,290	4,290	4,290	478
Adjusted R^2	0.606	0.305	0.604	0.307
ICD9 FEs	Yes	Yes	Yes	Yes
ICD9 Category × Year Trends	Yes	Yes	Yes	Yes

NOTE: Table presents coefficients from regressing the outcome in each column on an indicator for after the Medicare Memorandum of 2000 ("Post2000"), an indicator for whether a disease has an above median share of diagnoses among the elderly ("Medicare Share"), and the interaction between the two. We also include an indicator for after part D was announced in 2003 ("PostPartD") and the interaction of this with "MedicareShare". Observations are at the disease-year level. We control for ICD-9 fixed effects (which encompass the "Medicare Share" indicator) and the ICD-9 category by year trends. "Wald Test P-value" is the p-value comparing the $Post2000 \times MedicareShare$ coefficient in columns 2 vs. 3. Columns 1 and 4 are estimated using OLS. Columns 2 and 3 estimated using seemingly unrelated regressions. Standard errors are reported in parentheses, with * $p < 0.05$; ** $p < 0.01$ and *** $p < 0.001$.

Table A.7: IMPACT ON TRIALS, EXCLUDING FAST-TRACK DISEASES

	Ln(1 + # Total Trials) (1)	Ln(1 + # Trials Enrolling Only Age < 65) (2)	Ln(1 + # Trials Enrolling Any Age ≥ 65) (3)	Share of Participants Age ≥ 65 (4)
Post × MedicareShare	0.154** (0.0548)	-0.0216 (0.0227)	0.179*** (0.0509)	0.0393 (0.132)
Post	-0.138*** (0.0306)	-0.0231 (0.0160)	-0.143*** (0.0280)	0.0646 (0.111)
Mean of Dep. Var.	0.273	0.063	0.238	0.123
Diff. Wald Test p-value			0.00	
Observations	3,916	3,916	3,916	364
Adjusted R^2	0.534	0.27	0.53	0.198
ICD9 FEs	Yes	Yes	Yes	Yes
ICD9 Category × Year Trends	Yes	Yes	Yes	Yes

NOTE: Table presents coefficients from regressing the outcome in each column on an indicator for after the Medicare Memorandum of 2000 ("Post2000"), an indicator for whether a disease has an above median share of diagnoses among the elderly ("Medicare Share"), and the interaction between the two. Observations are at the disease-year level. ICD-9 codes associated with drugs receiving a Fast-Track designation are excluded. We also include an indicator for after part D was announced in 2003 ("PostPartD") and the interaction of this with "MedicareShare". We control for ICD-9 fixed effects (which encompass the "Medicare Share" indicator) and the ICD-9 category by year trends. "Wald Test P-value" is the p-value comparing the $Post2000 \times MedicareShare$ coefficient in columns 2 vs. 3. Columns 1 and 4 are estimated using OLS. Columns 2 and 3 estimated using seemingly unrelated regressions. Standard errors are reported in parentheses, with * $p < 0.05$; ** $p < 0.01$ and *** $p < 0.001$.

Table A.8: CHANGES IN AGE OF ENROLLED TRIAL PARTICIPANTS

	Trials Starting Pre-2000			Trials Starting Post-2000			Diff of Means P-value (7)
	Count (1)	Mean (2)	SD (3)	Count (4)	Mean (5)	SD (6)	
Mean Age							
All Diseases	88	48.26	17.24	772	51.73	15.79	.05
Young Diseases	22	38.5	11.06	201	40.99	16.91	.50
Old Diseases	66	51.51	17.76	571	55.51	13.49	.03
% of Enrollees Over 65							
All Diseases	43	.04	.16	1,061	.13	.24	.01
Young Diseases	20	0	.01	430	.04	.13	.19
Old Diseases	23	.07	.21	631	.2	.28	.03
Ln(No. of Enrollees Over 65 +1)							
All Diseases	64	.76	1.85	1,061	1.29	1.96	.03
Young Diseases	28	.02	.13	430	.53	1.29	.04
Old Diseases	36	1.33	2.32	631	1.81	2.15	.19
Ln(No. of Enrollees Under 65 +1)							
All Diseases	62	4.38	1.82	1053	4.36	1.83	.95
Young Diseases	28	4.73	1.42	427	4.86	1.73	.70
Old Diseases	34	4.09	2.07	626	4.03	1.82	.84

NOTE: This shows the mean age, percentage over 65, and count of the enrolled trial participants. Observations are at the trial-level. “All Diseases” refers to the full sample of trials. Due to the limited number of observations from missing trial enrollment data, we have expanded the sample to include all trials initiated between 1990 and 2005, excluding those from the year 2000. As in the main analysis, we restrict to Phase II and Phase III interventional trials with private sector funding and at least one location in the US. In instances where mean age was not available, median age was used instead. Column 7 shows p-values from a 2-sided t-test that compares columns 2 and 5.

Table A.9: IMPACT ON TRIAL OUTCOMES AND ADVERSE EVENTS

<i>A. Disease-Year Level Analysis, Impact on $\ln(\# \text{ Trials} + 1)$</i>			
	Trials With P-value < 0.05 (1)		Trials With P-value $\geq$ 0.05 (2)
Post $\times$ MedicareShare	0.0210 (0.0120)		0.0157** (0.00594)
Post	-0.0546*** (0.00914)		-0.0331*** (0.00571)
Mean of Dep. Var.	0.036		0.014
Diff. Wald Test p-value		0.65	
Observations	4,290		4,290
Adjusted R^2	0.178		0.121
ICD9 FEs	Yes		Yes
ICD9 Category $\times$ Year Trends	Yes		Yes
<i>B. Drug-Age Category-Gender-Year Level Analysis, Impact on Adverse Event Rate</i>			
	Elderly Disease (3)		Young Disease (4)
Post $\times$ Elderly	-0.161* (0.0681)		0.0828 (0.0490)
Post	0.122** (0.0453)		0.00379 (0.0364)
Mean of Dep. Var.	1.599		1.127
Diff. Wald Test p-value		0.00	
Observations	20,810		20,810
Gender FEs	Yes		Yes
ICD9 FEs	Yes		Yes

NOTE: Panel A presents coefficients from regressing the outcome in each column on an indicator for after the Medicare Memorandum of 2000 ("Post"), an indicator for whether a disease has an above median share of diagnoses among the elderly ("Medicare Share"), and the interaction between the two. Observations are at the disease-year level and we control for ICD-9 fixed effects (which encompass the "Medicare Share" indicator) and the ICD-9 category by year trends. Panel B presents coefficients from regressing the outcome in each column on an indicator for "Post", an indicator for whether the respondents are over 65 ("Elderly"), and the interaction between the two. Observations are at the drug by age category by sex by year level. The outcome is the number of adverse events reported per 1,000 prescriptions for either drugs in old diseases (column 3) or young diseases (column 4). We control for an indicator for elderly, gender, and ICD9 code. Results are estimated using seemingly unrelated regressions. Standard errors are reported in parentheses, with * $p < 0.05$; ** $p < 0.01$ and *** $p < 0.001$.